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# VHFACCaptureDefTopic

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About the VHFAC Capture

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About the VHFAC Capture

The VHFAC Capture instrument can accurately acquire differential signals with frequency components from DC to 300MHz. The range of user-programmed sample rates is 4ksps to 125Msps. The VHFAC Capture also has programmable voltage ranges and DC offset functionality for canceling out common mode or DC bias voltages.

This section includes the following topics:

- Features
- Safety Information

VHFAC Capture information includes the following sections:

- VHFAC Capture Theory of Operation
- VHFAC Capture Programming
- VHFAC Capture Debug Display
- VHFAC Capture Examples

Related Topics

- VHFAC Specifications
- VHFACCapture (VBT syntax)
- VHFAC DIB Design Guidelines

For more information on other VHFAC instruments, see:

- About the VHFAC Serial Bus
- About the VHFAC Source
- About the Time Measurement Unit

- About the VHFAC Trigger Control

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# VhfacCapture_about.1.2

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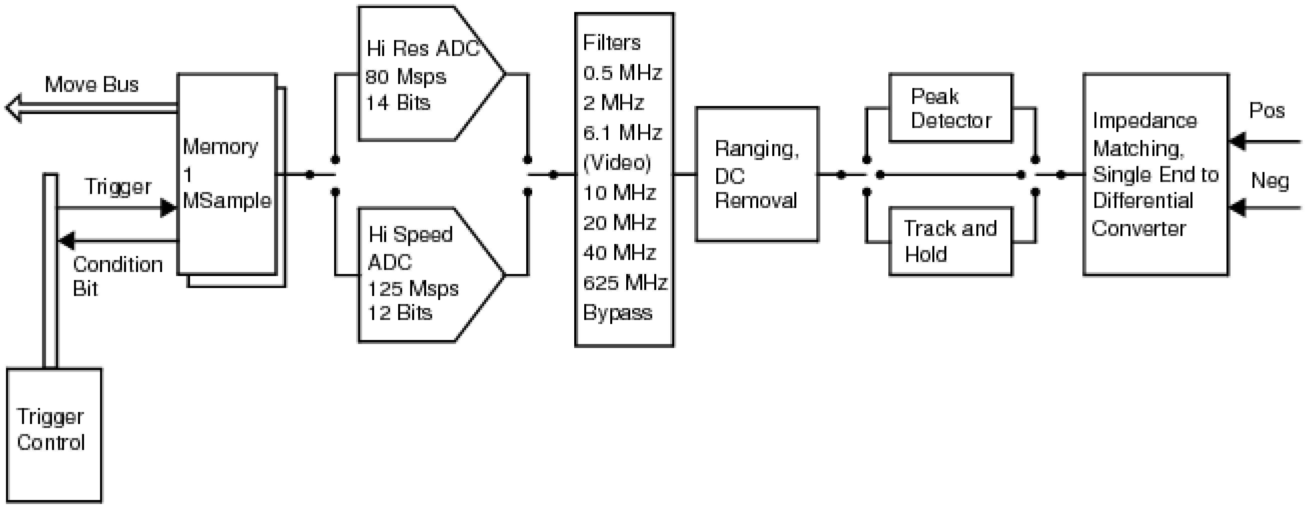
About the VHFAC Capture : Features

Features

The VHFAC Capture offers:

- Two independent capture instruments
- Selectable input ranges
- Programmable sample rates of 4ksps to 125Msps
- Programmable DC baseline using Common Mode Voltage
- Mixed-signal pattern microcode control
- Access to Pin Parametric Measurement Unit (PPMU) per pin
- Parameter sets (PSets) for pattern control of instrument settings
- DIB Access connectivity

As shown in the figure, the VHFAC Capture is a differential instrument that receives analog input signals on its two pins (Pos and Neg) from the DUT. The signal is passed through an optional coupling and offset stage, before the signal's amplitude is attenuated in the voltage ranging stage. A user-selectable analog anti-aliasing filter eliminates high frequency signal components. The signal is then passed through an analog-to-digital conversion process and is transformed to its digital equivalent. The data is then stored in the instrument's capture memory until the instrument moves the data to the Support Board with an instrument initiated move (IIM).




VHFAC Capture Instrument

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# VhfacCapture_about.1.3

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About the VHFAC Capture : Safety Information

## Safety Information

When operating the VHFAC Capture instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety for details.

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# VhfacCapture_debug.4.1

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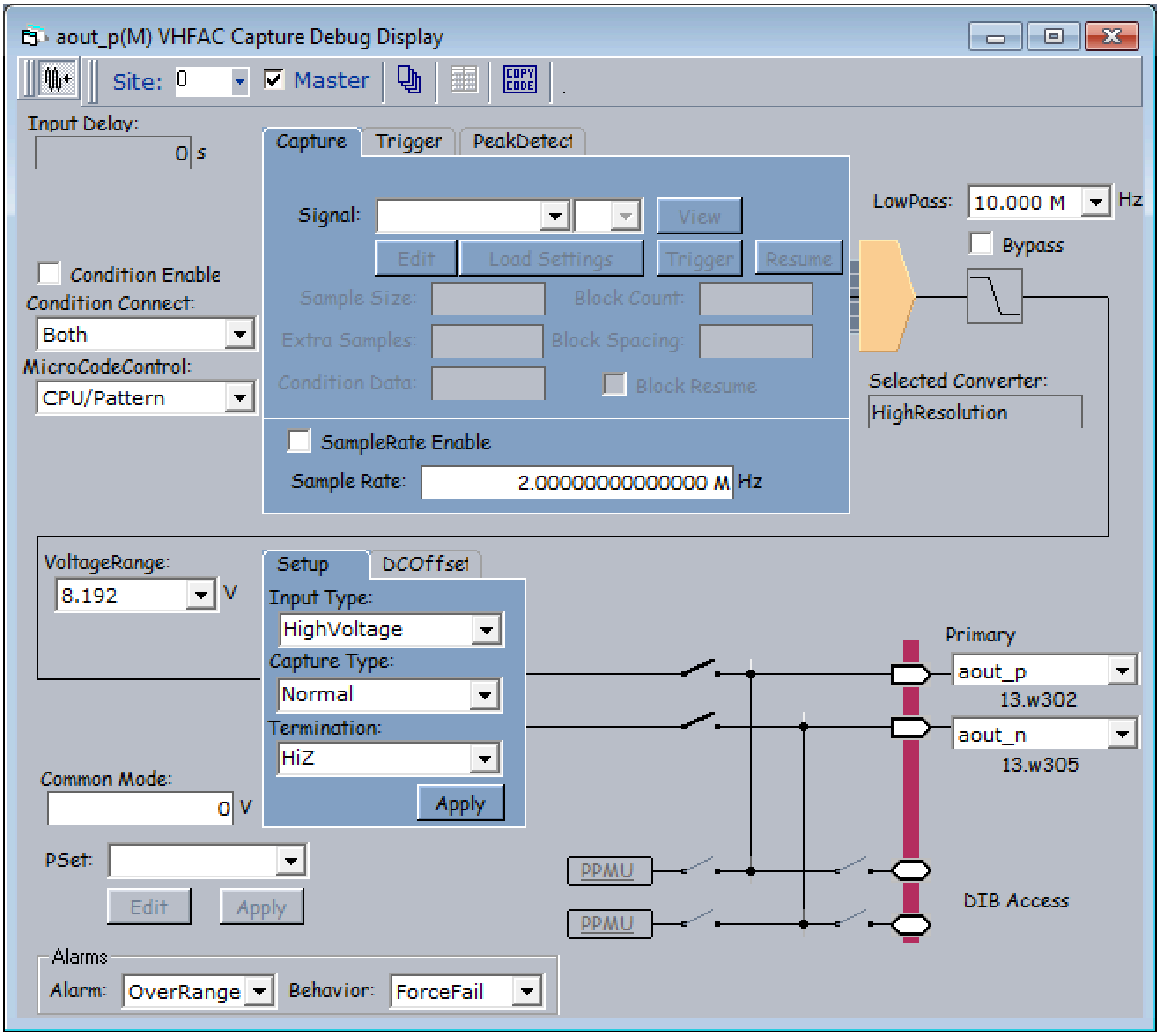
VHFAC Capture Debug Display

VHFAC Capture Debug Display

The VHFAC Capture Debug Display is an interactive tool that shows the state of the VHFAC Capture instrument in real time. Most of the parameters are interactive. Features that can be changed may be limited to a certain range or selection of options provided in a drop down menu. Non-interactive features are grayed out on the display and cannot be adjusted during debugging.




## VHFAC Capture Debug Display

This section describes features specific to the VHFAC Capture debug display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

## Starting the VHFAC Capture Debug Display

To launch the VHFAC Capture Debug Display, on the IG-XL Tools tab, in the Launch group, select Debug Displays>VHFAC Capture.

To start debugging, select a pin and a site to display the programmed values. Select sites and pins from the drop-down menu. Once a pin and a site are selected, parameters are shown for that instrument.

Without a test program loaded, the initial debug display produces and an error message:

! No pin exists of type VHFAC Capture

and the connection lines appear directly connected to the DUT (without relays, DIB Access, or PPMU instruments active). Load a validated test program and restart the debug display to remedy this situation.

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# VhfacCapture_debug.4.2

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons

Debug Display Parameters and Buttons

The VHFAC Capture Debug Display contains several sections. Each section has programmable features used to make adjustments to test parameters. The following topics are available:

- VHFAC Capture Schematic
- VHFAC Capture Alarms
- VHFAC Capture Tab
- VHFAC Capture Trigger Tab
- VHFAC Capture Peak Detect Tab
- VHFAC Capture TrigCmd Tab
- VHFAC Capture Pin Setup Information

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# VhfacCapture_debug.4.3

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture Schematic

VHFAC Capture Schematic

The VHFAC Capture Schematic section displays the current programmed settings of the selected VHFAC Capture instrument and allows you to change program parameters. Make connections to the DUT and other instruments by clicking the relay symbols.

The table below lists VHFAC Capture schematic debug display parameters and buttons.

VHFAC Capture Schematic Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Sample Rate | The number of samples to be captured per second.<br>• Minimum sample rate = 3.43 kHz<br>• Maximum sample rate = 125.00 MHz<br>The maximum value differs for each capture mode (determined by the combination of Input Type/Capture Type/Termination).<br>See VHFAC Capture Supported Modes. |
| Voltage Range | Sets the voltage range of the signal. A drop-down list provides the available voltage ranges: 8.0mV, 16.0mV, 32.0mV, 64.0mV, 128.0mV, 256.0mV, 512.0mV, 1.024V, 2.048V, 4.096V, and 8.192V Minimum and maximum values differ for each case of capture mode (Input Type/Capture Type/Termination). See VHFAC Capture Supported Modes. |

| | |
|---|---|
| LowPass | Lists the available low-pass filters: 500.0kHz, 2.0MHz, 6.1MHz, 10.0MHz, 20.0MHz, 40.0MHz, and 62.5MHz |
| Condition Enable | This checkbox enables the condition connect bits. |
| Condition Connect | The destination where the VHFAC Capture Instrument passes condition bits.<br>• Both<br>• Board<br>• PatGen |
| MicroCode Control | The means by which the microcode control executes:<br>• CPU/Pattern<br>• ExternalTrigger<br>• ExternalTrigger1<br>If ExternalTrigger or ExternalTrigger1 is selected, a TrigCmd tab appears to show the trigger commands for each capture start label. See VHFAC Capture TrigCmd Tab. |
| CommonMode | Sets the CommonMode Base Value.<br>• Minimum common mode voltage = -4.0 V<br>• Maximum common mode voltage = 4.0 V |
| Input Delay | Sets the time delay for the starting point of the waveform capture.<br>• Minimum input delay = 0.0 s<br>• Maximum input delay = 10.24 us |

| | |
|---|---|
| PSet | Selects a parameter set (PSet). Selecting a PSet from the list does not apply the settings to hardware. To apply the PSet to hardware, click Apply.<br>Edit brings up the PSet and Signals Editor (PSSE) with the selected PSet loaded. See PSet and Signals Editor in the TDE section. |

Setup Tab

Controls on the Setup tab allow you to set the VHFAC Capture input, capture type and termination. This combination of values determines sample rate and voltage range, as shown in the table below. Controls on this tab include the following:

VHFAC Capture Setup Tab Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Input Type | Selects single-ended or differential input. Differential input can be AC or DC-coupled. Choices are:<br>• ACDifferential<br>• DCDifferential<br>• SingleEnded<br>• HighVoltage<br>• Direct |
| Capture Type | Selects capture type. Choices are:<br>• Normal: Normal digitizing<br>• Undersampling: Sample and hold digitizing<br>• PeakDetect |

| | |
|---|---|
| Termination | Selects termination setting. Choices are:<br>• Term 50<br>• HiZ<br>• Voltage |

VHFAC Capture Supported Modes

| Capture Type | Input Type | Termination | DC Baseline Removal Debug Display Field | Sample Rate | Voltage Range |
|---|---|---|---|---|---|
| Normal | AC Differential | Term50 | None | 3.434k - 125M | 0.016 - 4.096[1] |
| | DC Differential | Term50 | DC Offset Normal Pos DC Offset Normal Pos | 3.434k - 125M | 0.008 - 4.096[1] |
| | | Voltage | Common Mode | 3.434k - 125M | 0.008 - 2.048[1] |
| | SingleEnded | Term50 | DC Offset Normal Single | 3.434k - 125M | 0.008 - 2.048 |
| | HighVoltage | Term50 | DC Offset Normal High Voltage | 3.434k - 80M | 0.008 - 8.192 |
| | | HiZ | DC Offset Normal High Voltage | 3.434k - 80M | 0.008 - 8.192 |
| | Direct | Term50 | None | 3.434k - 125M | 0.008 - 1.024 |
| Under-sampling | AC Differential | Term50 | None | 3.434k - 20M | 0.016 - 2.048[1] |
| | DC Differential | Term50 | DC Offset Normal Pos DC Offset Normal Pos | 3.434k - 20M | 0.008 - 2.048[1] |

| | | | | | |
|---|---|---|---|---|---|
| | | Voltage | Common Mode | 3.434k - 20M | 0.008 - 1.024[1] |
| | SingleEnded | Term50 | DC Offset Normal Single | 3.434k - 20M | 0.008 - 1.024 |
| | Direct | Term50 | None | 3.434k - 20M | 0.008 - 0.512 |
| PeakDetect | AC Differential | Term50 | None | 3.434k - 80M | 4.096[1] |
| | DC Differential | Term50 | None | 3.434k - 80M | 4.096[1] |
| | | Voltage | None | 3.434k - 80M | 2.048[1] |
| | SingleEnded | Term50 | None | 3.434k - 80M | 2.048 |
| | Direct | Term50 | None | 3.434k - 80M | 1.024 |

---

1 Differential voltage specifications assume a differential voltage across the two inputs. For example, in the 2.048 V range, the voltage at the high input can be 1.024 V peak and the voltage at the low input can be -1.024 V peak. This yields a differential voltage of 2.048 V across the high and low inputs.

DCOffset Tab

Controls on the DCOffset tab allow you to set single-ended DC Offset values for the VHFAC Capture. Controls on this tab include the following:

VHFAC Capture DCOffset Tab Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Normal | Sets the DC offset in normal mode.<br><br>• Minimum DC offset Normal = -2.048 V<br><br>• Maximum DC offset Normal = 2.048 V |

| | |
|---|---|
| HighVoltage | Sets the DC offset in high voltage mode.<br>• Minimum DC offset HighVoltage = -12.288 V<br>• Maximum DC offset HighVoltage = 12.282 V (When Input mode = HighVoltage, CaptureType = Normal, Termination = HiZ)<br>• Minimum DC offset HighVoltage = -8.192 V<br>• Maximum DC offset HighVoltage = 8.192 V |

For more information, see Pins.Offset in the VBT Syntax Reference.

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# VhfacCapture_debug.4.4

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture Alarms

VHFAC Capture Alarms

The alarm controls allow you to set behavior for one or all alarms to aid debugging. You can set the following parameters during debug:

VHFAC Capture Alarm Parameters and Buttons

| Parameter/Button | Description |
| --- | --- |
| Alarm | Selects the alarm for which to set the Behavior. The options are:<br>• AlarmAll<br>• OverRange<br>Selecting an alarm only affects the alarm for that pin. Note that AlarmAll and Over Range perform the same function. Also, the special option AlarmAll is write-only, so the behavior displayed for it is not updated if it is changed using VBT. |
| Behavior | Selects the action that should occur if the selected alarm condition exists for the pin. The options are:<br>• Continue<br>• Default |

- ForceBin
- ForceFail
- Off
- RaiseError

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# VhfacCapture_debug.4.5

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|---|---|

VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture Tab

VHFAC Capture Tab

Allows you to set and view sample size related parameters.

VHFAC Capture Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Signal | Lists the names of capture labels listed on the Wave Definitions sheet and created in user jobs. The state label under this drop-down list shows the current state (Stopped/Running) of the instrument. |
| View | This button is enabled when you select a signal containing an asterisk. Viewing a signal only makes a copy of the samples on the Support Board. The data is not removed from the Support Board. |
| Edit | Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section. |
| Load Settings | This button is enabled when a signal name is selected from the Signal field. It loads the settings of the specified Signal. That signal must be defined from VBT. |
| Trigger | Triggers the instrument to start the capture. |
| Resume | Triggers the instrument to resume capturing during block capture. |
| Sample Size | The number of samples to be captured. When a block capture is being performed, this parameter specifies the size of an individual |

| | |
|---|---|
| | block within that block capture. |
| Block Count | When performing a block capture, this field Indicates the number of blocks that will be captured. |
| Extra Samples | Specifies the number of additional samples to capture for both normal and block capture. In the case of block capture, the specified number of extra samples is captured for each block. |
| Block Spacing | When performing a block capture, this field indicates the number of ignored samples between blocks. Only applicable when “block resume mode” is false. |
| Block Resume | When performing a block capture, this checkbox indicates whether “block resume mode” is active. |
| Condition Data | Indicates the sample number when the condition bit goes high. |
| SampleRate Enable | Shows the state of SampleRate enable.<br>• Selected: SampleRate Enabled<br>• Not selected: SampleRate Disabled |
| Sample Rate | Represents the number of samples to be captured per second.<br>• Minimum sample rate = 3.43 kHz<br>• Maximum sample rate = 125.00 MHz |

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# VhfacCapture_debug.4.6

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture Trigger Tab

## VHFAC Capture Trigger Tab

Allows waveform captures to be made asynchronous to the test system. A change of state on the asynchronous trigger pin triggers a measurement on the pin.

VHFAC Capture Trigger Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Pin | Specifies the pin whose signal is used to asynchronously trigger the instrument. |
| ChanType | The asynchronous trigger pin’s channel type. The trigger channel type should match the asynchronous trigger pin channel type. |
| Which | A number identifying the trigger line used to asynchronously trigger the instrument. |

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# VhfacCapture_debug.4.7

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture Peak Detect Tab

VHFAC Capture Peak Detect Tab

Once the Peak Detect circuit is programmed, you can see the real time results in the VHFAC Capture Debug Display.

VHFAC Capture Peak Detect Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Peak Detect Value | Allows you to view real time peak-to-peak measurement results. |
| Measure | Allows you to perform a real time peak-to-peak measurement. |
| Reset | Allows you to perform a reset of the peak detect circuitry. |

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# VhfacCapture_debug.4.8

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture TrigCmd Tab

VHFAC Capture TrigCmd Tab

This tab frame shows the trigger commands for each capture start label.

VHFAC Capture Trigger Command Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Start Label | Select the start label to view its Trigger commands. |
| Command | What action will occur. |
| Label | What Signal it will occur on. |

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# VhfacCapture_debug.4.9

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VHFAC Capture Debug Display : Debug Display Parameters and Buttons : VHFAC Capture Pin Setup Information

VHFAC Capture Pin Setup Information

This information sets the pin and site to be debugged. Once a pin is selected the debug display contains the pin’s current setup.

VHFAC Capture Pin Setup Parameters and Buttons

| Parameter/Button | Description |
|---|---|
| Pin (Primary and Secondary) | Shows available pins from the pin map connected to the Capture instrument. Selecting a pin from the list loads parameters for that pin to be debugged. |
| Site | This drop-down box, located on the debug display tool bar, lists the sites for the selected pin that are available. Selecting a site loads the parameters for the selected pin. |
| Apply All Sites | This icon is on the debug display tool bar. Clicking this button writes all changes made to every active site. If the button is not depressed, only the current active site is affected by the changes. |
| PPMU | Click this button to open the PPMU debug display for the selected pin. See PPMU Debug Display. |

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# VhfacCapture_example.5.01

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VHFAC Capture Examples

VHFAC Capture Examples

This section contains program examples for the VHFAC Capture instrument. These IG-XL workbook examples show all IG-XL data and VBT code necessary for a complete running program.

The following topics are available:

- VHFAC Capture Basic Capture (VBT)
- VHFAC Capture Basic Capture (Pattern)

For more information, see:

- VHFACCapture (VBT Syntax Reference)
- VHFAC Capture Programming
- TTN: VHFAC Relay Usage and Life Improvement Guidelines

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# VhfacCapture_example.5.02

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VHFAC Capture Examples : VHFAC Capture Basic Capture (VBT)

VHFAC Capture Basic Capture (VBT)

This example uses the VHFAC Capture instrument to capture a signal.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- VBT Code
- DSP Code

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# VhfacCapture_example.5.03

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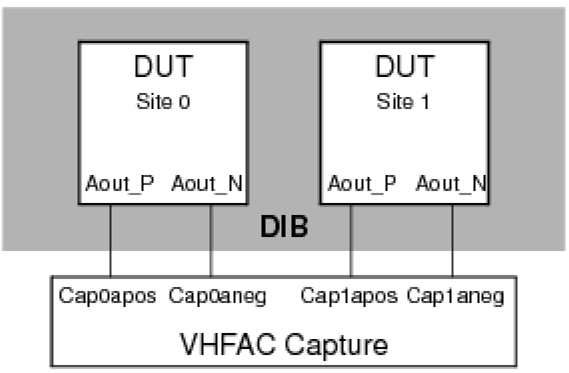
VHFAC Capture Examples : VHFAC Capture Basic Capture (VBT) : DUT Connections

## DUT Connections




VHFAC Capture DIB Connections

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# VhfacCapture_example.5.04

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VHFAC Capture Examples : VHFAC Capture Basic Capture (VBT) : DataTool Sheets

## DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Mixed Signal Timing Sheet

### Pin Map Sheet

**Pin Map**

| Group Name | Pin Name | Type |
|---|---|---|
| | Aout_P | Analog |
| | Aout_N | Analog |

### Channel Map Sheet

# Channel Map

**DIB ID:** 

**View Mode:** Signal

| Device Under Test | | Tester Channel | | |
|---|---|---|---|---|
| Pin Name | Package Pin | Type | Site 0 | Site 1 |
| Aout_P | | VHFACCapture | 13.cap0apos | 13.cap1apos |
| Aout_N | | VHFACCapture | 13.cap0aneg | 13.cap1aneg |

## Test Instances Sheet

# Test Instances

| | Test Procedure | | | Sheet |
|---|---|---|---|---|
| Test Name | Type | Name | Called As | Mixed Signal Timing |
| **Cap125MHzFs** | VBT | VHFACCaptureSetupTrig_VBT | | Cap125MHz |

The test instance named Cap125MHzFs has the following test-specific parameters as shown in the Instance Editor:

Instance: Cap125MHzFs

Parameters | Timing and Levels | Limits

| Name | Value |
|---|---|
| CapPin | Aout_P |
| CapSigName | Cap125MHzFs |
| CapVrange | 1.024 |

## Flow Table Sheet

# Flow Table

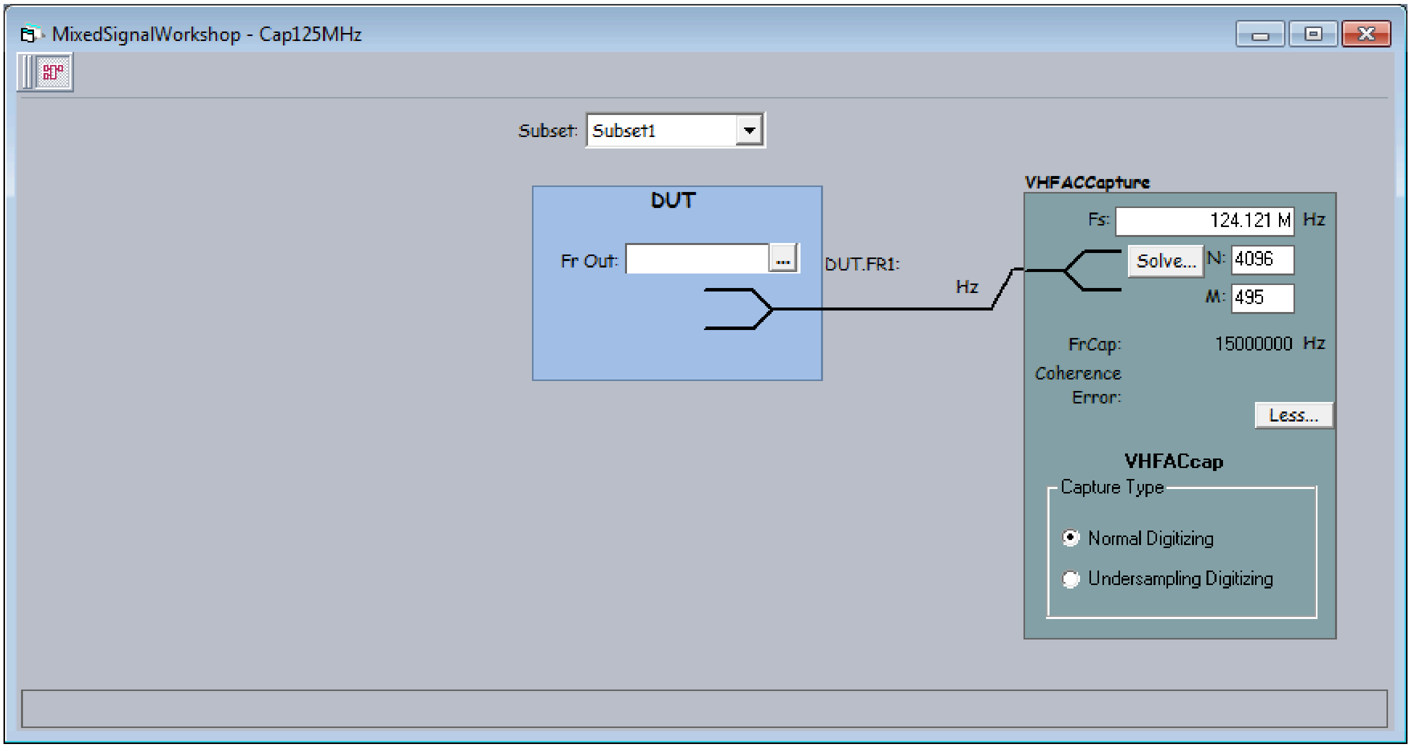
Flow Domain: 

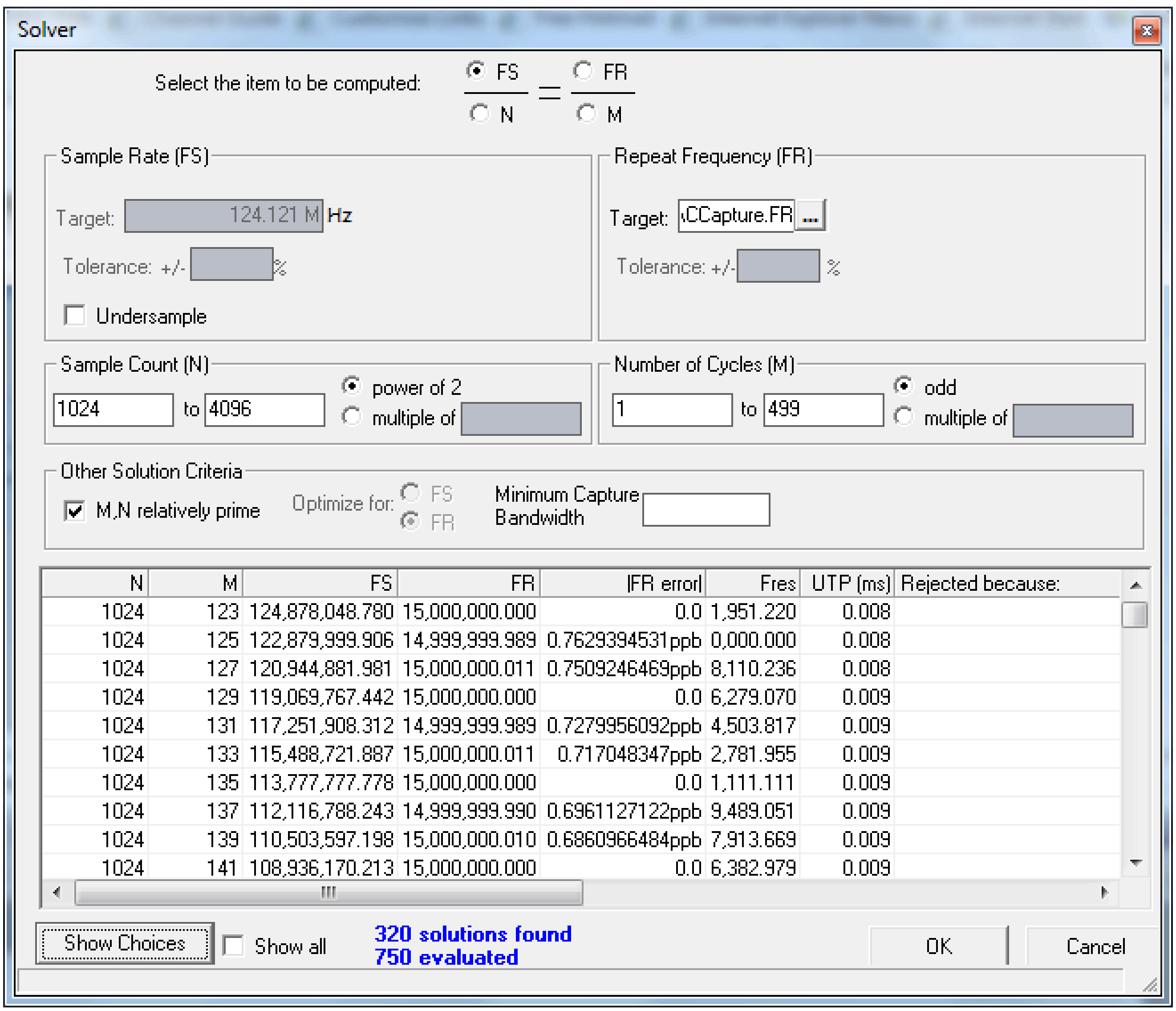
| | | Gate | | | Command | | | | Limits | | Datalog Display | | | Bin Number | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Label | Enable | Job | Part | Env | Opcode | Parameter | TName | TNum | LoLim | HiLim | Scale | Units | Format | Pass | Fail |
| | | | | | **Test-defer-limi** | Cap125MHzFs | | 1000 | | | | | | | |
| | | | | | Use-Limit | Cap125MHzFs | Freq | 1000 | 0 | 1E+08 | | | | 1 | 9 |
| | | | | | **limits-all** | | | | | | | | | | |

## Mixed Signal Timing Sheet

# Mixed Signal Timing

| | | Resource | | Clocking | | | | | Instrument | Waveform | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Name | Subset | Type | ID | Fs | N | Fr | M | USR | Data | Definition | Filter |
| **Cap125MHz** | Subset1 | VHFACCapture | | 124121212.1 | 4096 | 15000000 | 495 | 1 | DIGMODE=NORMAL | | |


MixedSignalWorkshop - Cap125MHz
Subset: Subset1
DUT
Fr Out:
DUT.FR1:
Hz
VHFACCapture
Fs: 124.121 M Hz
Solve...
N: 4096
M: 495
FrCap: 15000000 Hz
Coherence Error:
Less...
VHFACcap
Capture Type
Normal Digitizing
Undersampling Digitizing


Solver
Select the item to be computed:
FS
FR
N
M
Sample Rate (FS)
Target: 124.121 M Hz
Tolerance: +/- %
Undersample
Repeat Frequency (FR)
Target: CCapture.FR
Tolerance: +/- %
Sample Count (N)
1024 to 4096
power of 2
multiple of
Number of Cycles (M)
1 to 499
odd
multiple of
Other Solution Criteria
M,N relatively prime
Optimize for: FS FR
Minimum Capture Bandwidth
N M FS FR |FR error| Fres UTP (ms) Rejected because:
1024 123 124,878,048.780 15,000,000.000 0.0 1,951.220 0.008
1024 125 122,879,999.906 14,999,999.989 0.7629394531ppb 0,000.000 0.008
1024 127 120,944,881.981 15,000,000.011 0.7509246469ppb 8,110.236 0.008
1024 129 119,069,767.442 15,000,000.000 0.0 6,279.070 0.009
1024 131 117,251,908.312 14,999,999.989 0.7279956092ppb 4,503.817 0.009
1024 133 115,488,721.887 15,000,000.011 0.717048347ppb 2,781.955 0.009
1024 135 113,777,777.778 15,000,000.000 0.0 1,111.111 0.009
1024 137 112,116,788.243 14,999,999.990 0.6961127122ppb 9,489.051 0.009
1024 139 110,503,597.198 15,000,000.010 0.6860966484ppb 7,913.669 0.009
1024 141 108,936,170.213 15,000,000.000 0.0 6,382.979 0.009
Show Choices
Show all
320 solutions found
750 evaluated
OK
Cancel

| < | > |
|---|---|

# VhfacCapture_example.5.05

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| --- | --- |

VHFAC Capture Examples : VHFAC Capture Basic Capture (VBT) : VBT Code

VBT Code

```
Public Function VHFACCaptureSetupTrig_VBT(CapPin As PinList, CapSigName As String, CapVrange As Double, _
                        Optional TNames_ As String = "Freq") As Long
    On Error GoTo errHandler
    Dim CapWave As New DSPWave
    Dim Freq As New SiteDouble
    ' Connect VHFAC Capture differentially to the DIB
    TheHdw.VHFACCapture.Pins(CapPin).Connect tlVHFACCaptureConnectPrimaryPos + tlVHFACCaptureConnectPrimaryNeg, tlVHFACCaptureConnectToDUT
    ' Set up the VHFAC Capture Signal
    TheHdw.VHFACCapture.Pins(CapPin).Signals.Add CapSigName
    With TheHdw.VHFACCapture.Pins(CapPin).Signals(CapSigName)
        .ApplyMixedSignalTiming
        .LowPass.Bypass = False
        .LowPass.Value = .SampleRate / 2 ' >= Nyquist bandwidth
        .VoltageRange = CapVrange
        .InputType = tlVHFACCaptureInputTypeDCDifferential
        .CaptureType = tlVHFACCaptureSignalsCaptureTypeNormal
        .Termination = tlVHFACCaptureTerminationTerm50
        .LoadSettings
    End With
    ' Trigger the Capture
    TheHdw.VHFACCapture.Pins(CapPin).Signals(CapSigName).Trigger
```

```
    ' Bind the capture to a DSP Wave
    CapWave = TheHdw.VHFACCapture.Pins(CapPin).Signals(CapSigName).DSPWave
    ' Process the captured waveform
    rundsp.CalcFreq CapWave, Freq
    ' Compare results to limits
    TheExec.Flow.TestLimit resultVal:=Freq, unit:=unitHz, ForceResults:=tlForceFlow
    ' Wait for capture to complete
    Dim CapDone As Boolean
    CapDone = TheHdw.VHFACCapture.Pins(CapPin).Signals.WaitForCaptureDone
    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

---

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# VhfacCapture_example.5.06

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VHFAC Capture Examples : VHFAC Capture Basic Capture (VBT) : DSP Code

## DSP Code

```
Public Function CalcFreq(ByVal CapWave As DSPWave, ByRef Freq As Double) As Long
    Freq = CapWave.Spectrum.CalcFrequencyFromSpectrum
End Function
```

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# VhfacCapture_example.5.07

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VHFAC Capture Examples : VHFAC Capture Basic Capture (Pattern)

VHFAC Capture Basic Capture (Pattern)

This example uses the VHFAC Capture instrument to capture a signal.

This workbook example includes the following program components:

- DUT Connections
- DataTool Sheets
- Patterns
- VBT Code
- DSP Code

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# VhfacCapture_example.5.08

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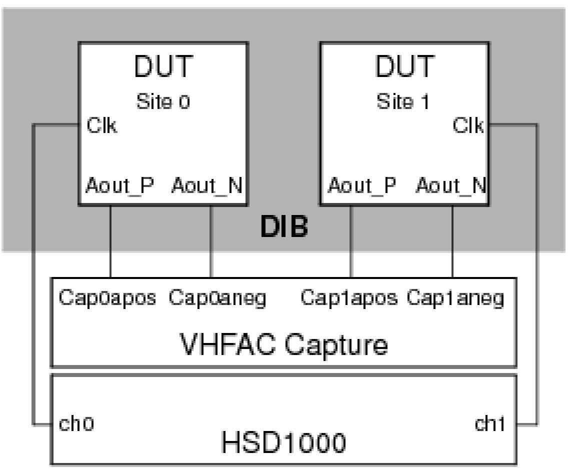
VHFAC Capture Examples : VHFAC Capture Basic Capture (Pattern) : DUT Connections

## DUT Connections




VHFAC Capture DIB Connections

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# VhfacCapture_example.5.09

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VHFAC Capture Examples : VHFAC Capture Basic Capture (Pattern) : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet
- Mixed Signal Timing Sheet
- Time Sets (Basic)
- Pin Levels Sheet

Pin Map Sheet

# Pin Map

| Group Name | Pin Name | Type |
|---|---|---|
| | Aout_P | Analog |
| | Aout_N | Analog |
| | Clk | Input |

## Channel Map Sheet

# Channel Map

**DIB ID:** 

**View Mode:** Signal

| Device Under Test | | Tester Channel | | |
|---|---|---|---|---|
| Pin Name | Package Pin | Type | Site 0 | Site 1 |
| Aout_P | | VHFACCapture | 13.cap0apos | 13.cap1apos |
| Aout_N | | VHFACCapture | 13.cap0aneg | 13.cap1aneg |
| Clk | | I/O | 16.ch0 | 16.ch1 |

## Test Instances Sheet

# Test Instances

TERADYNE

| | Test Procedure | | | DC Specs | | AC Specs | | Sheet Parameters | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Test Name | Type | Name | Called As | Category | Selector | Category | Selector | Time Sets | Edge Sets | Pin Levels | Mixed Signal Timing |
| CapAmpSteps125MHzFs | VBT | VHFACCaptureSetupTrig_POP | | | | | | TSB | | Levels | Cap125MHz |

The test instance named Cap125MHzFs has the following test-specific parameters and values as shown in the Instance Editor:

Instance: CapAmpSteps125MHzFs

Parameters | Timing and Levels | Limits

| Name | Value |
|---|---|
| CapPin | Aout_P |
| CapSigName | Cap125MHzFs1V |
| CapVrange | 1.024 |
| patName | .\VHFACCapture_POP.P |

## Flow Table Sheet

# Flow Table

TERADYNE

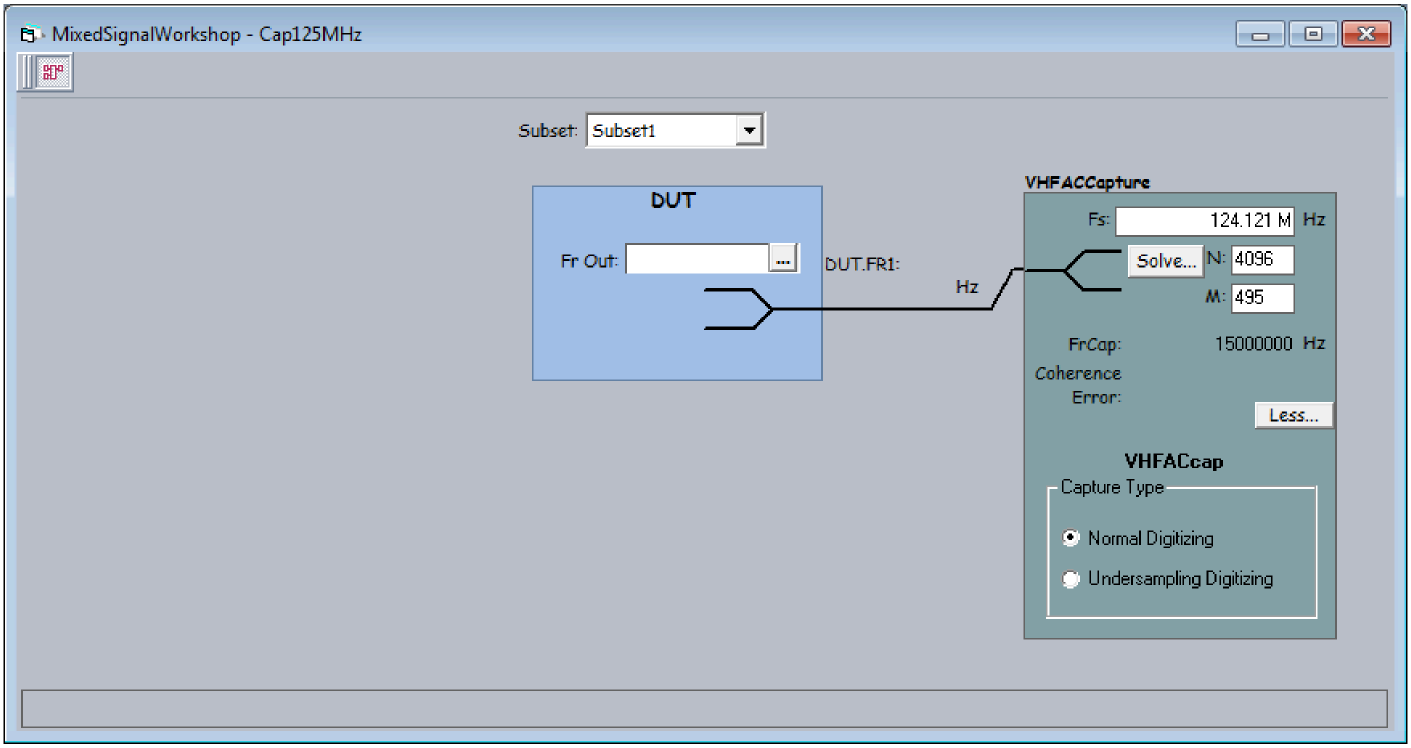
**Flow Domain:** 

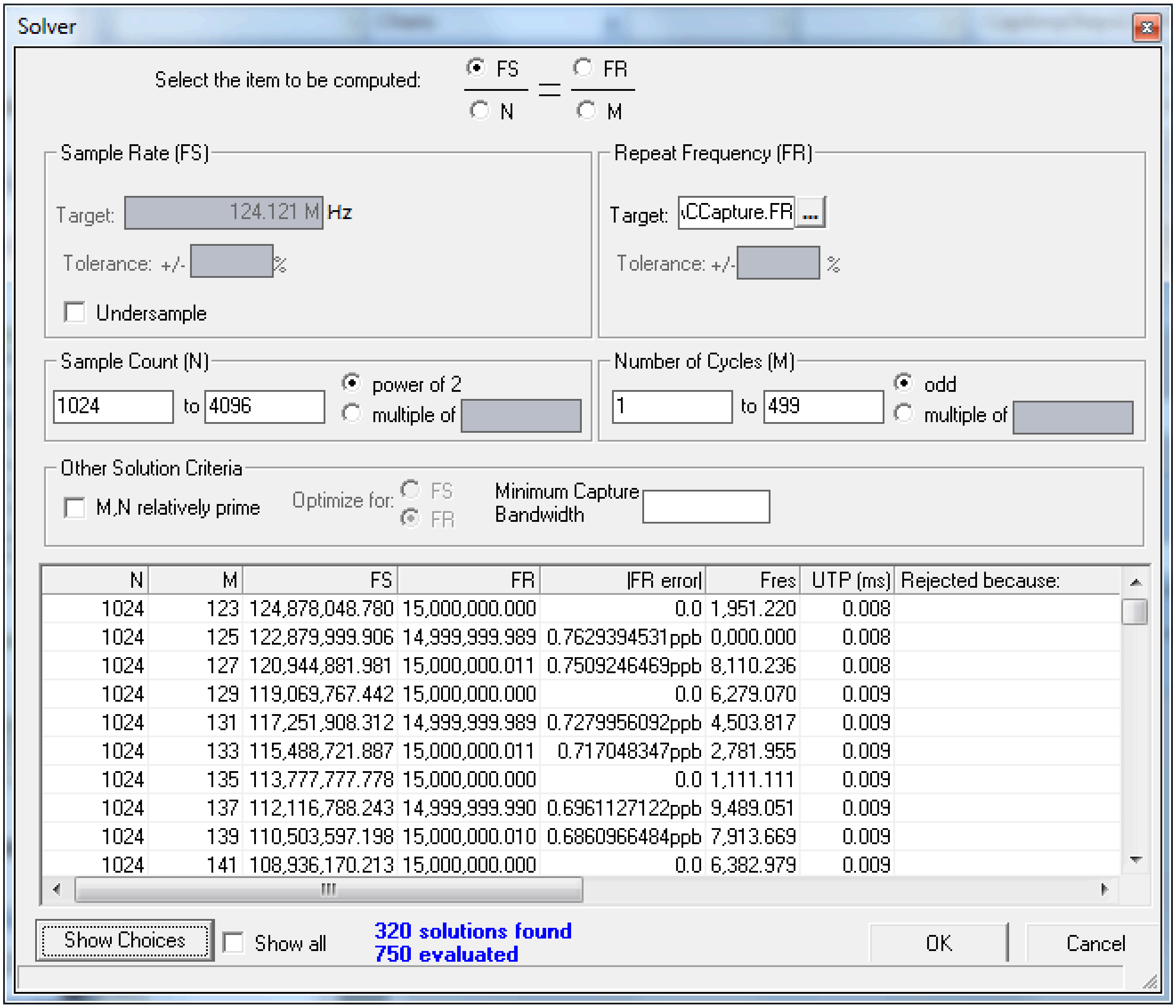
| | | Gate | | | Command | | | | Limits | | Datalog Display | | | Bin Number | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Label | Enable | Job | Part | Env | Opcode | Parameter | TName | TNum | LoLim | HiLim | Scale | Units | Format | Pass | Fail |
| | | | | | **Test-defer-limits** | CapAmpSteps125MHzFs | | 2000 | | | | | | | |
| | | | | | Use-Limit | CapAmpSteps125MHzFs | Amp1 | 2000 | 0 | 1.1 | | | | 1 | 7 |
| | | | | | Use-Limit | CapAmpSteps125MHzFs | Amp2 | 2001 | 0 | 0.55 | | | | 1 | 7 |
| | | | | | Use-Limit | CapAmpSteps125MHzFs | Amp3 | 2002 | 0 | 0.3 | | | | 1 | 7 |
| | | | | | **limits-all** | | | | | | | | | | |

## Mixed Signal Timing Sheet

# Mixed Signal Timing

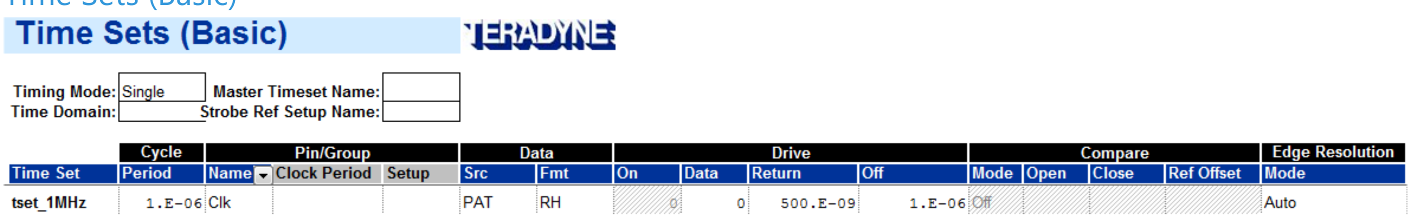
| | | Resource | | Clocking | | | | | Instrument | Waveform | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Set Name | Subset | Type | ID | Fs | N | Fr | M | USR | Data | Definition | Filter |
| Cap125MHz | Subset1 | VHFACCapture | | 124121212.1 | 4096 | 15000000 | 495 | 1 | DIGMODE=NORMAL | | |






| N | M | FS | FR | \|FR error\| | Fres | UTP (ms) | Rejected because: |
|---|---|---|---|---|---|---|---|
| 1024 | 123 | 124,878,048.780 | 15,000,000.000 | 0.0 | 1,951.220 | 0.008 | |
| 1024 | 125 | 122,879,999.906 | 14,999,999.989 | 0.7629394531ppb | 0,000.000 | 0.008 | |
| 1024 | 127 | 120,944,881.981 | 15,000,000.011 | 0.7509246469ppb | 8,110.236 | 0.008 | |
| 1024 | 129 | 119,069,767.442 | 15,000,000.000 | 0.0 | 6,279.070 | 0.009 | |
| 1024 | 131 | 117,251,908.312 | 14,999,999.989 | 0.7279956092ppb | 4,503.817 | 0.009 | |
| 1024 | 133 | 115,488,721.887 | 15,000,000.011 | 0.717048347ppb | 2,781.955 | 0.009 | |
| 1024 | 135 | 113,777,777.778 | 15,000,000.000 | 0.0 | 1,111.111 | 0.009 | |
| 1024 | 137 | 112,116,788.243 | 14,999,999.990 | 0.6961127122ppb | 9,489.051 | 0.009 | |
| 1024 | 139 | 110,503,597.198 | 15,000,000.010 | 0.6860966484ppb | 7,913.669 | 0.009 | |
| 1024 | 141 | 108,936,170.213 | 15,000,000.000 | 0.0 | 6,382.979 | 0.009 | |

## Time Sets (Basic)



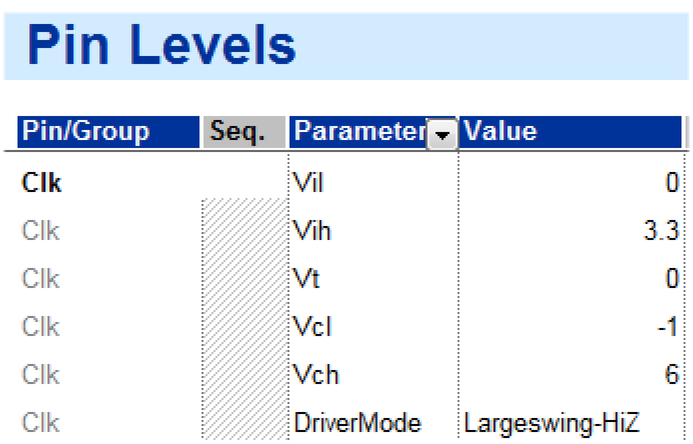
**Time Sets (Basic)**

TERADYNE

**Timing Mode:** Single **Master Timeset Name:**
**Time Domain:** **Strobe Ref Setup Name:**

| | Cycle | Pin/Group | | | Data | | Drive | | | | Compare | | | | Edge Resolution |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Time Set | Period | Name | Clock Period | Setup | Src | Fmt | On | Data | Return | Off | Mode | Open | Close | Ref Offset | Mode |
| tset_1MHz | 1.E-06 | Clk | | | PAT | RH | 0 | 0 | 500.E-09 | 1.E-06 | Off | | | | Auto |

## Pin Levels Sheet



**Pin Levels**

| Pin/Group | Seq. | Parameter | Value |
|---|---|---|---|
| **Clk** | | Vil | 0 |
| Clk | | Vih | 3.3 |
| Clk | | Vt | 0 |
| Clk | | Vcl | -1 |
| Clk | | Vch | 6 |
| Clk | | DriverMode | Largeswing-HiZ |

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# VhfacCapture_example.5.10

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VHFAC Capture Examples : VHFAC Capture Basic Capture (Pattern) : Patterns

## Patterns

| TSet | Command | Label | Vector | Aout_P (VHFACCapture) | Clk | Comment |
|---|---|---|---|---|---|---|
| tset_1MHz | | | 0 | Enable_Inst_Cond | 0 | enable condition bit, use for branch… |
| tset_1MHz | branch_expr = (!inst_cond) | | 1 | | 0 | |
| tset_1MHz | | | 2 | Trig | 0 | trigger Capture #1 |
| tset_1MHz | | capture_loop1: | 3 | | 0 | wait for Capture #1 to complete |
| tset_1MHz | if (branch_expr) jump capture_loop1 | | 4 | | 0 | |
| tset_1MHz | | | 5 | PSet 500mV | 0 | apply PSET for 500mV test |
| tset_1MHz | repeat 5000 | | 6 | | 0 | wait 5ms for PSET to settle |
| tset_1MHz | branch_expr = (!inst_cond) | | 7 | | 0 | |
| tset_1MHz | | | 8 | Trig | 0 | trigger Capture #2 |
| tset_1MHz | | capture_loop2: | 9 | | 0 | wait for Capture #2 to complete |
| tset_1MHz | if (branch_expr) jump capture_loop2 | | 10 | | 0 | |
| tset_1MHz | | | 11 | PSet 250mV | 0 | apply PSET for 250mV test |
| tset_1MHz | repeat 5000 | | 12 | | 0 | wait 5ms for PSET to settle |
| tset_1MHz | branch_expr = (!inst_cond) | | 13 | | 0 | |
| tset_1MHz | | | 14 | Trig | 0 | trigger Capture #3 |
| tset_1MHz | | capture_loop3: | 15 | | 0 | wait for Capture #3 to complete |
| tset_1MHz | if (branch_expr) jump capture_loop3 | | 16 | | 0 | |
| tset_1MHz | | | 17 | Disable_Inst_Cond | 0 | disable condition bit |
| tset_1MHz | halt | | 18 | | 0 | |

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# VhfacCapture_example.5.11

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VHFAC Capture Examples : VHFAC Capture Basic Capture (Pattern) : VBT Code

VBT Code

Option Explicit

```
Public Function VHFACCaptureSetupTrig_POP(CapPin As PinList, CapSigName As String, _
    CapVrange As Double, patName As Pattern, _
    Optional TNames_ As String = "Amp1|Amp2|Amp3") As Long

    On Error GoTo errHandler

    Dim CapWave As New DSPWave
    Dim Amp1 As New SiteDouble
    Dim Amp2 As New SiteDouble
    Dim Amp3 As New SiteDouble

    ' Apply digital levels and timing to the hardware
    Call TheHdw.Digital.ApplyLevelsTiming(False, True, True, tlPowered)

    ' Connect VHFAC Capture differentially to the DIB
    TheHdw.VHFACCapture.Pins(CapPin).Connect tlVHFACCaptureConnectPrimaryPos + _
    tlVHFACCaptureConnectPrimaryNeg, tlVHFACCaptureConnectToDUT

    ' Set up the VHFAC Capture Signal
    TheHdw.VHFACCapture.Pins(CapPin).Signals.Add CapSigName

    With TheHdw.VHFACCapture.Pins(CapPin).Signals(CapSigName)
```

```
        .ApplyMixedSignalTiming

        .LowPass.Bypass = False
        .LowPass.Value = .SampleRate / 2 ' >= Nyquist bandwidth

        .VoltageRange = CapVrange
        .InputType = tlVHFACCaptureInputTypeDCDifferential
        .CaptureType = tlVHFACCaptureSignalsCaptureTypeNormal
        .Termination = tlVHFACCaptureTerminationTerm50

        .LoadSettings
    End With

    TheHdw.VHFACCapture.Pins(CapPin).Signals.DefaultSignal = CapSigName

    ' Set up the VHFAC Capture PSets
    If TheExec.ExecutionCount = 0 Then ' Set up PSets on first run of program only
        With TheHdw.VHFACCapture.Pins(CapPin).PSets
            .Add "500mV"
            .Add "250mV"
        End With

        With TheHdw.VHFACCapture.Pins(CapPin)
            .PSets("500mV").VoltageRange = 0.512
            .PSets("500mV").LowPass.Bypass = .Signals(CapSigName).LowPass.Bypass
            .PSets("500mV").LowPass.Value = .Signals(CapSigName).LowPass.Value
            .PSets("500mV").CommonMode = .Signals(CapSigName).CommonMode
            .PSets("500mV").Offset.HighVoltage =  _
                .Signals(CapSigName).Offset.HighVoltage
            .PSets("500mV").Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCPos) =
           .Signals(CapSigName).Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCPos)
            .PSets("500mV").Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCNeg) =
.Signals(CapSigName).Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCNeg)
            .PSets("500mV").Offset.Normal(tlVHFACCaptureNormalDCOffsetSingleEnded) =
.Signals(CapSigName).Offset.Normal(tlVHFACCaptureNormalDCOffsetSingleEnded)
            .PSets("500mV").PeakDetect.StartAddress = 0
            .PSets("500mV").SelectChannel = tlVHFACCaptureSelectChannelPrimary
```

```
            .PSets("250mV").VoltageRange = 0.256
            .PSets("250mV").LowPass.Bypass = .Signals(CapSigName).LowPass.Bypass
            .PSets("250mV").LowPass.Value = .Signals(CapSigName).LowPass.Value
            .PSets("250mV").CommonMode = .Signals(CapSigName).CommonMode
            .PSets("250mV").Offset.HighVoltage = _
                .Signals(CapSigName).Offset.HighVoltage
            .PSets("250mV").Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCPos) =
.Signals(CapSigName).Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCPos)
            .PSets("250mV").Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCNeg) =
.Signals(CapSigName).Offset.Normal(tlVHFACCaptureNormalDCOffsetDiffDCNeg)
            .PSets("250mV").Offset.Normal(tlVHFACCaptureNormalDCOffsetSingleEnded) =
.Signals(CapSigName).Offset.Normal(tlVHFACCaptureNormalDCOffsetSingleEnded)
            .PSets("250mV").PeakDetect.StartAddress = 0
            .PSets("250mV").SelectChannel = tlVHFACCaptureSelectChannelPrimary
        End With
    End If

    ' Start the pattern
    With TheHdw.Patterns(patName)
        .Load
        .Start
    End With

    ' Bind the capture to a DSPWave
    CapWave = TheHdw.VHFACCapture.Pins(CapPin).Signals(CapSigName).DSPWave

    ' Process the captured waveform
    rundsp.CalcAmplitudes CapWave, Amp1, Amp2, Amp3

    ' Compare results to limits
    TheExec.Flow.TestLimit resultVal:=Amp1, unit:=unitVolt, _
    ForceResults:=tlForceFlow
    TheExec.Flow.TestLimit resultVal:=Amp2, unit:=unitVolt, _
    ForceResults:=tlForceFlow
    TheExec.Flow.TestLimit resultVal:=Amp3, unit:=unitVolt, _
    ForceResults:=tlForceFlow

    ' Wait for pattern to halt
```

```
    TheHdw.Digital.Patgen.HaltWait

    Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
```

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# VhfacCapture_example.5.12

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VHFAC Capture Examples : VHFAC Capture Basic Capture (Pattern) : DSP Code

DSP Code

```
Public Function CalcAmplitudes(ByVal CapWave As DSPWave, ByRef Amp1 As Double, _
    ByRef Amp2 As Double, ByRef Amp3 As Double) As Long

    Amp1 = CapWave.Spectrum.CalcAmplitudeFromSpectrum
    CapWave.Next
    Amp2 = CapWave.Spectrum.CalcAmplitudeFromSpectrum
    CapWave.Next
    Amp3 = CapWave.Spectrum.CalcAmplitudeFromSpectrum

End Function
```

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# VhfacCapture_prog.3.01

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VHFAC Capture Programming

VHFAC Capture Programming

This section describes programming of a VHFAC Capture to capture a waveform from a device.

The following topics are available:

- Simulated Configuration Files
- Programming Overview
- VHFAC Capture DataTool Programming
- VHFAC Capture Pattern Control
- Additional Programming Features

For more information, see:

- VHFACCapture (VBT Syntax Reference)
- VHFAC Capture Examples

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# VhfacCapture_prog.3.02

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VHFAC Capture Programming : Simulated Configuration Files

Simulated Configuration Files

You can use the VHFAC Capture offline by forcing IG-XL to map the VHFAC. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

The following SimulatedConfig.txt file maps the VHFAC board in slot 13:

```
<Version>
1
</Version>

<TEST_HEAD TH 1>
#slot[.subslot]   Type              idprom (type serial rev company)

13            VHFAC             805-014-00 0000000 0528-A 5445
              VHFAC             939-376-00 0000000 0528-A 5445
              VHFACCapture0         939-244-00 0000000 0450-B 5445
              VHFACTime           939-376-00 0000000 0528-A 5445
              VHFACSource5          939-247-10 0000000 0450-B 5445
              VHFACSource4          939-247-10 0000000 0450-B 5445
              VHFACCapture1         939-244-00 0000000 0450-B 5445
```

24.0 SupportBoard 974-220-12 0000000 0000-A 0000
SupportBoard 956-354-00 0000000 0000-A 0000

62.0 DistributionBoard 956-355-00 0000000 0000-A 0000
DistributionBoard 956-355-00 0000000 0000-A 0000

</TEST_HEAD TH 1>

To use the VHFAC Capture offline, the SimulatedConfig.txt file must include every board needed in the IG-XL workbook. The SimulatedConfig.txt file represents the state of the offline tester. To use the VHFAC Capture online, you need not edit any files. IG-XL automatically maps all boards in the tester.

For more information, see Tester Configuration.

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# VhfacCapture_prog.3.03

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VHFAC Capture Programming : Programming Overview

Programming Overview

To develop a test program that uses the VHFAC Capture instrument, follow this basic procedure:

1. Create a pin map with pins of the DUT (Programming the Pin Map)
2. Create a channel map that allocates the tester resources to the DUT pins (Programming the Channel Map).
3. Create a mixed signal timing set (Creating a Timing Set Using the Mixed Signal Workshop).
4. Create a test flow, the order in which the test will execute, in the Flow Table sheet.
   - Create a test function using VBT code in the Visual Basic editor.
   - Instantiate the VBT test and set the test values in the Instance Editor.
5. Validate and run the test program.

VBT syntax is described in the VHFACCapture section of the VBT Syntax Reference.

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# VhfacCapture_prog.3.04

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VHFAC Capture Programming : VHFAC Capture DataTool Programming

VHFAC Capture DataTool Programming

This section describes how to program several DataTool worksheets to handle basic VHFAC Capture parameters. It includes the following topics:

- Programming the Pin Map
- Programming the Channel Map
- Creating a Timing Set Using the Mixed Signal Workshop

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# VhfacCapture_prog.3.05

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VHFAC Capture Programming : VHFAC Capture DataTool Programming : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. For each pin requiring a VHFAC Capture resource, select Analog as the pin type from the Type column drop-down list. Analog is the only valid pin type for VHFAC Capture pins.

☑ **Note:**

If DIB Access or Multiple Resources are used for a device pin and the VHFAC Capture is not the primary resource, then you can select a different type for the device pin.

For more information, see Pin Map Sheet in the DataTool section.

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# VhfacCapture_prog.3.06

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VHFAC Capture Programming : VHFAC Capture DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. For the VHFAC Capture instrument, you must select a channel type from the Type column menu for each pin or pin group. Valid VHFAC Capture channel types are:

| | |
|---|---|
| VHFACCapturePrimaryNeg | VHFAC Source Primary Negative Pin |
| VHFACCapturePrimaryPos | VHFAC Source Primary Positive Pin |
| VHFACCaptureSecondaryNeg | VHFAC Source Secondary Negative Pin |
| VHFACCaptureSecondaryPos | VHFAC Source Secondary Positive Pin |

Type the tester channel information value for each site to connect a DUT pin to the instrument.This table lists the available VHFAC Capture channels. Note that in the table, x indicates the DIB slot number of the VHFAC Capture.

VHFAC Capture Channel Resources

| VHFAC Capture Pogo Pin | DIB Channel | Signal Name |
|---|---|---|
| VHFAC Capture0 Primary Pos | x.w302 | x.cap0apos |
| VHFAC Capture0 Primary Neg | x.w305 | x.cap0aneg |
| VHFAC Capture0 Secondary Pos | x.w308 | x.cap0bpos |
| VHFAC Capture0 Secondary Neg | x.w311 | x.cap0bneg |

| | | |
|---|---|---|
| VHFAC Capture1 Primary Pos | x.m302 | x.cap1apos |
| VHFAC Capture1 Primary Neg | x.m305 | x.cap1aneg |
| VHFAC Capture1 Secondary Pos | x.m308 | x.cap1bpos |
| VHFAC Capture1 Secondary Neg | x.m311 | x.cap1bneg |

For more information, see Channel Map Sheet in the DataTool section.

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# VhfacCapture_prog.3.07

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VHFAC Capture Programming : VHFAC Capture DataTool Programming : Creating a Timing Set Using the Mixed Signal Workshop

Creating a Timing Set Using the Mixed Signal Workshop

The major function of the VHFAC Capture instrument is to capture analog waveforms from the DUT. The VHFAC Capture instrument uses timing sets to specify key capture sampling parameters such as sample rate and sample size.

The Mixed Signal Workshop (MSW) tool helps you to build reusable mixed signal timing solutions that meet coherence requirements. The output of the MSW is a mixed signal timing set which contains key capture sampling parameters such as sample rate and sample size. Timing sets are stored on the Mixed Signal Timing sheet in the IG-XL workbook and become a reusable timing solution for future tests.

To create the required timing set:

1. Insert and set up a Mixed Signal Timing sheet in the IG-XL workbook.

    a. Select Insert>Insert IG-XL Worksheet in the Sheet group on the IG-XL Main tab.

    b. Select Mixed Signal Timing from the Sheet type drop-down list in the Insert IG-XL Worksheet dialog box.

    c. Type a name for the sheet in the Sheet name box (optional). Default name is Mixed Signal Timing.

    d. Click OK.

e. Type a timing set name in the Set Name column and press Enter.

f. Right-click the timing set name and select Edit from the context menu to start the Mixed Signal Workshop (MSW) tool in TDE.

The MSW display appears and displays the DUT and a capture resource block.

2. Add a VHFAC Capture resource to the MSW display, if necessary.

a. Select MSW>Add Capture.

b. Select VHFAC Capture from the Select Resource dialog box.
Leave the Resource ID box empty, unless this timing set is using additional VHFAC Capture instruments. If that is the case, assign the current capture instrument a unique ID number to distinguish it from the others.

c. Click the OK in the Select Resource dialog box.
The VHFAC Capture resource block is added to the MSW display.

☑ **Note:**
MSW launches a blank set with a default Capture instrument resource. If you do not require this resource, select the block, then right-click. Choose Delete instrument from the menu.

3. Specify the Capture timing. Typically, the objective of programming the VHFAC Capture instrument is to find a sample rate (Fs) and sample size (N) that provide the required bandwidth, frequency resolution, and if necessary, signal coherence.

a. Click the Solve button in the Capture resource block to start the Solver tool.

b. Select Fs to solve for the sample rate (Fs).

c. In the Other Solution Criteria frame, type in a value for the minimum capture bandwidth of interest. Typically, for sine wave testing, you will also select M,N Relatively Prime.

d. In the Number of Cycles frame, specify a range for the bin of the fundamental frequency. Typically, for sine wave testing, choose the number of cycles (M) to be odd to facilitate finding a timing solution that has M and N mutually prime. This step is optional.

e. In the Sample Count frame, specify a range for the capture's sample size. Typically, for sine wave testing, choose the number of samples (N) to be a power of two to facilitate finding a timing solution that has M and N mutually prime.

f. In the Repeat Frequency frame, specify the expected capture frequency (FR).

☑ **Note:**
In many cases, the capture timing solution (Fs and N) constrains the value for a coherent source frequency (FR). The description above (step 3) assumes that this is not the case and that the capture instrument is being programmed to coherently capture a known source frequency.

g. Click Show Choices, select a solution from the table, and click OK.
The best solution can depend on the specifics of your test's needs. Note that if no solutions are found, you must modify the inputs to the Solver. Typically, this is sample size range or cycle count range that was used. Additionally, you can select Show All to see why some of the solutions were not valid.

4. Save the timing set by clicking the Save button in the toolbar.

For more information, see Mixed Signal Workshop.

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# VhfacCapture_prog.3.08

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VHFAC Capture Programming : VHFAC Capture Pattern Control

VHFAC Capture Pattern Control

The VHFAC Capture instrument can be controlled by patterns and microcode allowing changes to instrument states and parameters during a program. Repeatable timing can be obtained by programming an instrument with microcodes, patterns, and parameter sets (PSets).

A pattern is created independently of the test program, by using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well as defining the pattern microcode used for controlling pattern execution.

The two forms of pattern control are:

| | |
|---|---|
| Microcode | A microcode controls a single function, such as starting or stopping the capturing of a waveform. |
| PSets | A VHFAC Capture PSet contains parameters such as voltage range and DC Offset. When an instrument receives a PSet from the pattern, it goes to the programmed state represented by that PSet. |

The following topics are available:

- VHFAC Capture Pattern Microcodes
- PSets

Creating a Pattern for Use in a Test Instance

You can create a VHFAC Capture pattern by using the PatternTool within TDE or by writing the pattern as an ASCII text file. For small patterns, use the PatternTool. Note that you must first have a valid pin map. Then:

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| 1. | Start the PatternTool by selecting TDE>Tools>Pattern Tool from the menu bar in the TDE window. |
| 2. | Open a window for a new pattern by clicking File>New. |
| 3. | Select single or dual mode and other options on the Options worksheet. |
| 4. | Read in the pin map by clicking File>Read Pinmap. |
| 5. | Create a Pin List on the Pin List sheet. |
| 6. | Create Pin Setups and define instruments on the Pin Setups and Instruments worksheets if required. |
| 7. | Import Tsets and Labels. |
| 8. | Add Labels, Opcodes, and Vectors on the Vectors worksheet. |

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VHFAC Capture Programming : VHFAC Capture Pattern Control : VHFAC Capture Pattern Microcodes

VHFAC Capture Pattern Microcodes

A series of extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a waveform is captured. To establish repeatable timing or to synchronize with other instruments (such as HSD) in the tester, use a pattern. If you want to start capturing a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the microcodes that are responsible for issuing commands to VHFAC Capture instrument from a pattern.

VHFAC Capture Pattern Microcodes

| Microcode | Description |
|---|---|
| Trig <name> | Begins capturing under the specified signal label. |
| Resume | When in BlockResume mode, continues a capture that has been suspended. |
| Resync | Initiates a sequence within the instrument that synchronizes the instrument clock. |
| Measure_Peak | Captures the peak detector output and stores it into Peak Capture memory. |
| Reset_Peak | Resets the peak detector. |
| PSet <name> | Applies a PSet of the specified name. |
| Enable_Alarm | Enables all VHFAC Capture alarms. |

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|---|---|
| Disable_Alarm | Disables the VHFAC Capture from asserting the instrument condition. |
| Enable_Inst_Cond | Enables the VHFAC Capture to drive the condition bit. |
| Disable_Inst_Cond | Disables the VHFAC Capture from driving the condition bit. |

VHFAC Capture Microcode Rules

When using capture microcodes, follow these rules:

- When issuing multiple VHFAC Capture Trigger microcode commands in a pattern (this includes Trig, Resume, and Measure_Peak), the spacing between successive Trigger commands must obey the following rule. Failure to follow this rule will result in a “Microcode Tossing Error”:
  - The time between “Trigger” microcode commands must be greater than 1/Source Sample Rate.
- If a Trig or Resume microcode is issued while the instrument is writing to CMEM, a “Memory in Use” error will occur.
- When issuing a Resync microcode, the time between the Resync and any other microcode must be greater than 6.5 ms to allow for the source channel clock to come into phase alignment with the HSD T0 clock.
- When issuing a PSet microcode, the time between the PSet and any other microcode must be greater than 5 ms to allow for hardware settling.
- For Enable_Inst_Cond and Disable_Inst_Cond microcode commands, there is no limitation for the spacing between successive microcodes.

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VHFAC Capture Programming : VHFAC Capture Pattern Control : PSets

PSets

A parameter set (PSet) is a set of values that modifies instrument hardware setting using a pattern microcode. PSets allow the test program to change instrument setups without stopping and later resuming pattern execution. PSets improve repeatability because they operate independent of the host computer. This also increases throughput by reducing the amount of handshaking required between the pattern and the host computer. Although PSets are a hardware capability that changes the setup of an instrument during a pattern burst, they are defined in the VBT language and invoked using PSet microcode.

PSet properties and methods for the VHFAC Capture instrument are described in the following sections. Note that PSets do not support incremental programming and that only a subset of instrument parameters can be controlled by PSets.

For more information, see VHFACCapture in the VBT Syntax Reference.

Switching Between Input Paths Using PSets

The VHFAC Capture is equipped with a mux input path separated into Primary and Secondary Inputs. A feature exists which allows you to switch between the Primary and Secondary Inputs on the fly. This ability is useful for instance when performing an I&Q measurement using a “hot-switch” technique. This technique allows you to accurately measure phase difference using a single VHFAC Capture instrument and capturing I&Q sequentially. (The I signal is connected to Primary Input, and the Q signal is connected to the Secondary Input.)

The VHFAC Capture supports this hot-switch technique in the pattern. While some connections are initially set up (and finally restored) in VBT, the ultimate switching between primary and secondary is accomplished in the pattern using a PSet. You accomplish the setup using PreConnect syntax. Connection cleanup is accomplished using the PostDisconnect and ResetSelectChannel syntax. The switch between inputs is accomplished by applying a PSet containing the SelectChannel syntax.

Adding a VHFAC Capture PSet to a Test Program

To add VHFAC Capture PSet function to a test program do the following:

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| 1. | Use the Add statement to add a specific name to the instrument instance. For example, to add a new PSet "a" to all the instrument instances behind pin "Aout", the code would look like this: |

```
TheHdw.VHFACCapture.Pins("Aout").PSets.Add ("a")
```

| | |
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| 2. | Use the Set statement to set all PSet parameters. |

Set all the parameters of PSet "a" (Vrange = 2.048, DC Offset Single Ended = 1.1V, DC Offset POS = 0.0V, DC Offset NEG = 0.0V, DC Offset High Voltage = 0.0V, DC Offset Common Mode = 0.0V, Select Channel = "Primary", Peak Detect Start Address = 0, Filter Bypass = False, Filter Value = 2MHz), with the following code:

```
Call TheHdw.VHFACCapture.Pins("Aout").PSets("a").Set(2.048, 1.1, 0#, 0#, 0#, 0#, _
        tlVHFACCaptureSelectChannelPrimary, 0, False, 2000000#)
```

| | |
|---|---|
| 3. | Use the Apply statement to apply the PSet to the indicated pins. In the example, you would apply PSet "a" from VBT as follows: |

```
TheHdw.VHFACCapture.Pins("Aout").PSets("a").Apply
```

☑ **Note:**

On applying a PSet, a hardware settling time of > 5 ms is required. This settle time is not managed by IG-XL. You must manage this yourself.

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VHFAC Capture Programming : Additional Programming Features

## Additional Programming Features

VHFAC Capture instruments include several additional features that are used for specific purposes. These include:

- VHFAC Capture Condition Bit
- Block Capture
- VHFAC Capture Peak Detect

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VHFAC Capture Programming : Additional Programming Features : VHFAC Capture Condition Bit

VHFAC Capture Condition Bit

The VHFAC capture instrument can generate a condition bit that synchronizes with an analog waveform in the capture memory. The condition bit is distributed to other instruments on the VHFAC board using the Trigger Control instrument or the condition bit can be sent directly to the PatGen.

The capture condition bit is a digital signal programmed to "Start" (go "High") with a specific sample of the captured signal.

The condition bit is routed with VBT commands. To route to the board for use with the Trigger Control (that is, to distribute the condition signal to other VHFAC instruments) or the PatGen use the following code:

TheHdw.VHFACCapture.Pins(SrcPin).Condition.Connect tlVHFACCaptureConditionToBoard

To route to the PatGen, to be used for conditional looping/branching within a pattern, use the following code:

TheHdw.VHFACCapture.Pins(SrcPin).Condition.Connect tlVHFACCaptureConditionToPatgen

To route to both the VHFAC board and PatGen simultaneously, use the following code:

TheHdw.VHFACCapture.Pins(SrcPin).Condition.Connect tlVHFACCaptureConditionToBoth

VBT Programming Example

The following code shows how to program the capture condition bits:

```
With TheHdw.VHFACCapture.Pins(CapPin)
    .Signals(CapSig).ConditionData.Value = 1024
    .Condition.Connect tlVHFACCaptureConditionToBoth
    .Condition.Enable = True
End With
```

In this example, the condition data value is set to 1024. This is the sample number that starts (sets high) the condition bit. The condition connect is set to both, sending the condition data to the board (other VHFAC instruments) and the pattern generator. Condition enable is set to true, although this can also be done from the pattern.

For more information, see:

- TheHdw.VHFACCapture.Pins.Pins.Condition
- TheHdw.VHFACCapture.Pins.Signals.Item.ConditionData

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VHFAC Capture Programming : Additional Programming Features : Block Capture

Block Capture

The VHFAC capture instrument has a block capture feature which allows you to capture specific pieces of data (rather than data containing the desired data as well as data which will be discarded) and store only that data into capture memory. This feature is valuable for conserving capture memory and reducing the DSP processing time that would normally be required to select the captured data of interest.

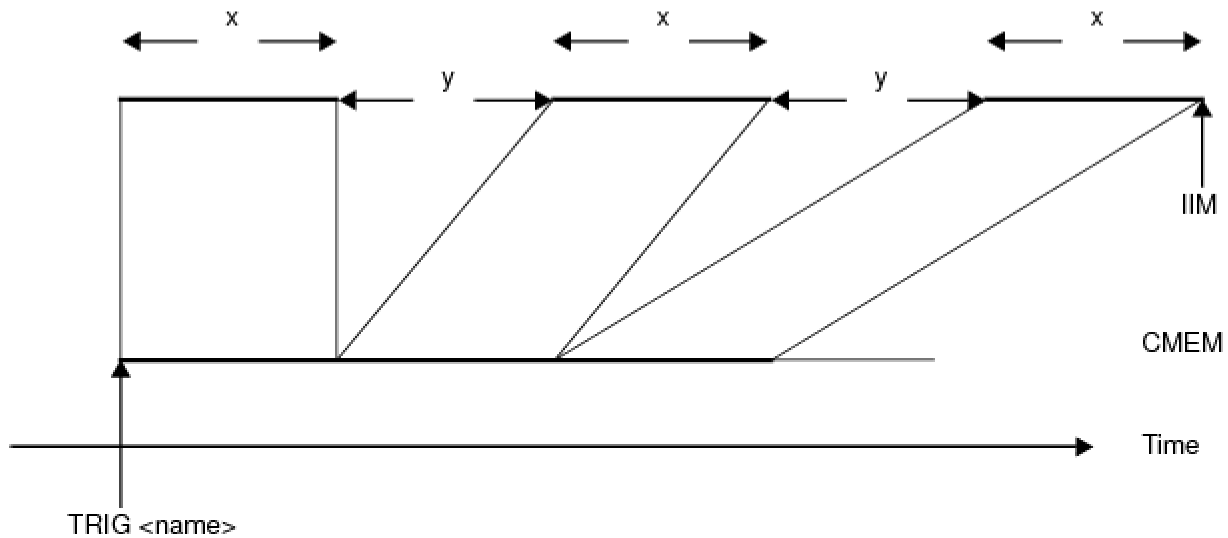
The two modes of Block Capture are:

- Normal Block Capture
- Block Resume

Normal Block Capture

For normal block capture mode, you must specify a BlockSize, BlockSpacing, and BlockCount.

In the following figure, BlockSize is set to x and BlockSpacing is set to y. The size of the capture in capture memory will be 3x, instead of 3x+2y. The blocks are concatenated together in memory and are not moved until the total sample size has been captured.




Normal Block Capture

You must indicate the beginning of the first block using a TRIG command (either from VBT or the pattern).

VBT Programming Example

The following code shows how to program a normal block capture:

```
' Set up the VHFAC Capture Signal
   TheHdw.VHFACCapture.Pins(CapPin).Signals.Add ("CapSig1")
With TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig1")
      .SampleRate = 80000000#
      .BlockResume = False    ' Mode of Block Capture
      .SampleSize = 1024      ' Size of each "Block"
      .BlockCount = 16        ' Number of "Blocks" to capture
      .BlockSpacing = 512     ' Fixed space between each "Block"
      .LoadSettings
   End With
' Trigger the Capture
   TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig1").Trigger
```

In this example, BlockResume mode is set to False. When set to False, a normal block capture is programmed. If set to True, a resume block capture is specified. This mode is optional for normal block captures as False is the default value of this parameter.

BlockSize is programmed using the SampleSize parameter.

The BlockCount parameter programs the number of blocks to capture.

The BlockSpacing parameter programs the amount of fixed space (as measured in samples) between blocks. In this example, BlockSpacing is programmed to allow 512 samples to pass before the next block is captured.

Finally, note that the line:

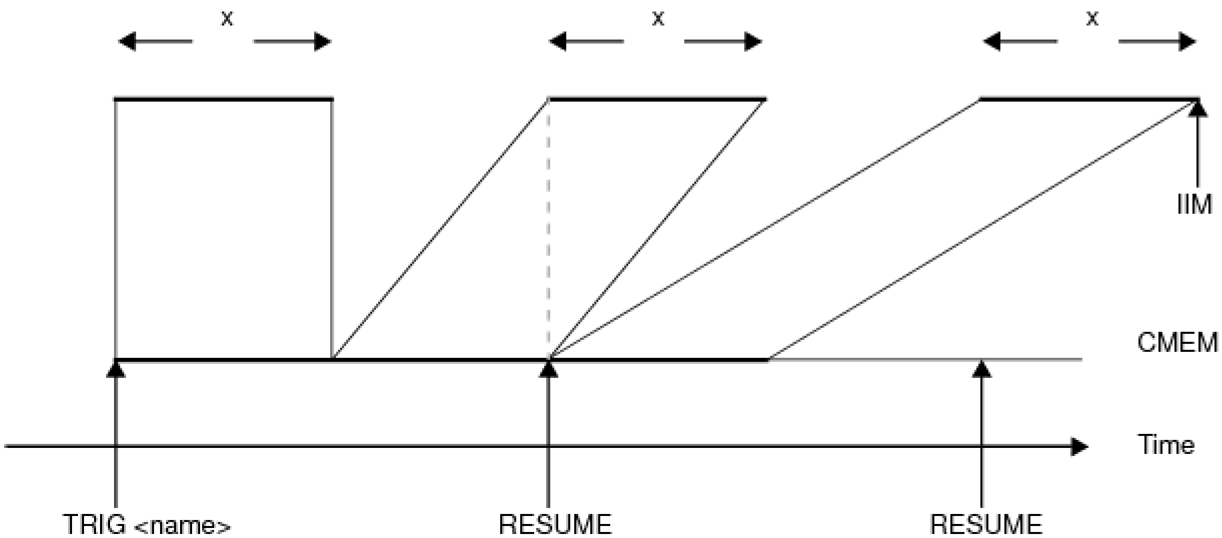
TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig1").Trigger triggers the instrument, and the specified block capture will occur with no further user intervention required.

Block Resume

For the Block Resume Capture mode, you must specify a BlockSize and BlockCount. BlockSpacing will depend on when the RESUMEs are executed.

You must indicate the beginning of the first block with a TRIG command (either from VBT or the pattern) and the beginning of each successive block with a RESUME command (either from VBT or the pattern).

In the following figure, BlockSize has been set to x, and BlockSpacing is variable.




Block Resume Block Capture

VBT Programming Example

The following code shows how to program a block resume block capture:

```
' Set up the VHFAC Capture Signal
    TheHdw.VHFACCapture.Pins(CapPin).Signals.Add ("CapSig2")
    With TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig2")
        .SampleRate = 80000000#
        .BlockResume = True      ' Mode of Block Capture
        .SampleSize = 4096       ' Size of each "Block"
        .BlockCount = 2          ' Number of "Blocks" to capture
        .LoadSettings
    End With
    ' Trigger the Capture
    TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig2").Trigger
    TheHdw.Wait (0.005)
    ' Resume the next block capture
    TheHdw.VHFACCapture.Pins(CapPin).Signals.Resume
```

The BlockResume mode is set to True. This is how IG-XL differentiates between normal and resume block captures.

In this example, the BlockSize has been programmed using the SampleSize parameter. And BlockCount has been set to 2 to specify the number of blocks to capture.

The line:

TheHdw.VHFACCapture.Pins(CapPin).Signals("CapSig2").Trigger

is needed to trigger the instrument to capture the first specified block. Whereas the line:

TheHdw.VHFACCapture.Pins(CapPin).Signals.Resume

is needed to resume the capture until all blocks (that is, the number specified in the BlockCount) are captured.

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VHFAC Capture Programming : Additional Programming Features : VHFAC Capture Peak Detect

VHFAC Capture Peak Detect

The VHFAC Capture instrument is equipped with additional circuitry, referred to as the Peak Detector, that enables fast, peak-to-peak measurements of analog waveforms. More specifically, the Peak Detector is a Comparator Feedback type detector. This type of detector is a real peak-to-peak detector that is not affected by the crest factor of the input wave.

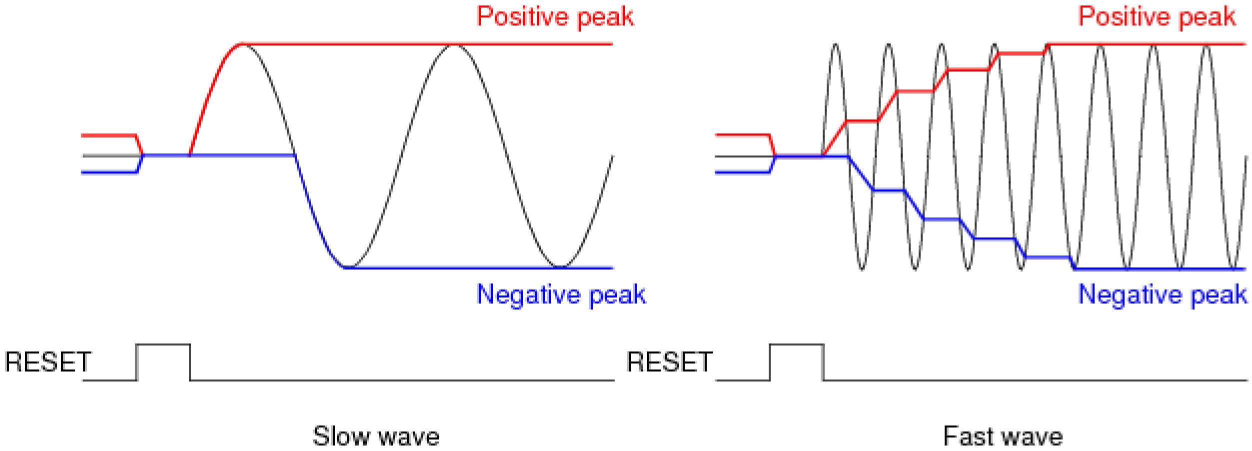
Measurements made using the Peak Detector are performed through the normal input DIB Pogo pins of the VHFAC Capture instrument.

Theory of Operation

Before making measurements with the Peak Detector, the circuit must be reset using a RESET command using either pattern microcode or VBT syntax. When RESET is high, the Peak Detector capacitors are discharged. When RESET is low, the peak-to-peak detector is activated.

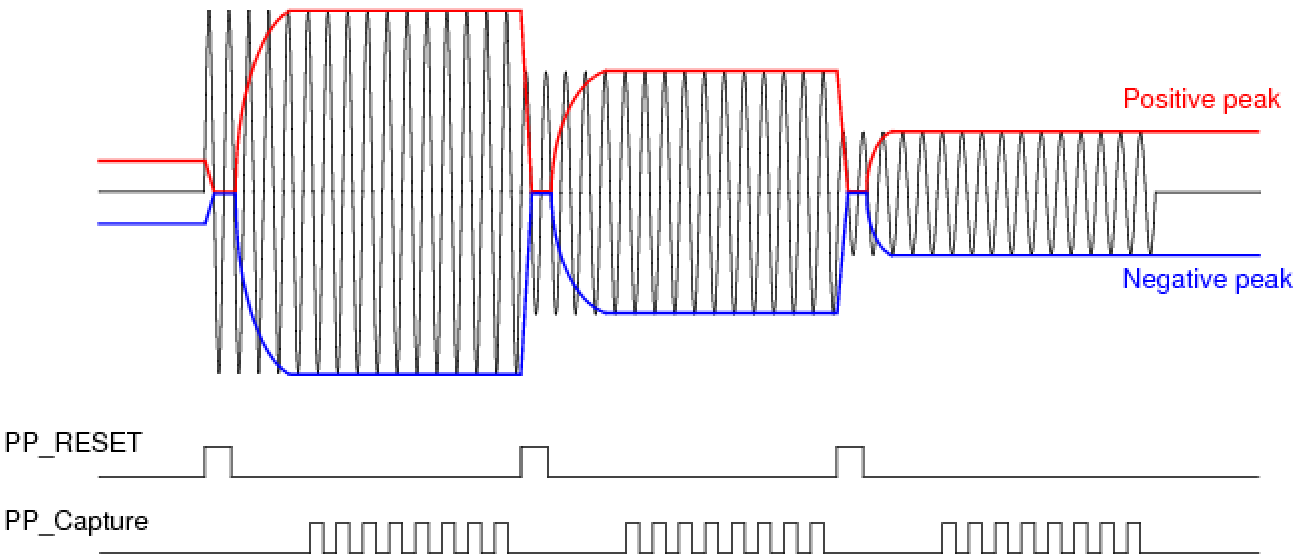
At first, both capacitors are at the low level after being activated. When an input signal is applied and crosses over the capacitor level, comparator and translator output become high and the capacitors are charged. The charge continues until the capacitor levels reach the peak level of the input wave. If the wave is slower than the slew rate of Peak Detector, the peak-to-peak level is reached after one cycle of the waveform. If the wave is faster than the slew rate, multiple cycles are required to reach the peak level of the waveform as shown in the figure.

For information on slew rate and other instrument characteristics, see VHFAC Capture Instrument (VHFACCapture) in the VHFAC instrument specifications.




Reaching Peak Level of a Waveform with Slow and Fast Waves

A combination of the RESET and MEASURE commands will allow you to measure the envelope of a waveform. Also, multiple MEASUREs allow averaging to capture more stable measurements. Finally, multiple RESETs and MEASUREs allow you to perform averaged measurements on multiple envelopes as shown in the figure below.




Measuring Multiple Envelopes

VBT Code Example

You can use VBT code to set up the peak detector circuit and make the desired number of peak measurements. An example of this is shown below.

```
' Set up the Peak Detector
 With TheHdw.VHFACCapture.Pins(CapPin)
    Call .SetMode(tlVHFACCaptureCaptureTypePeakDetect, _
            tlVHFACCaptureInputModeSingleEnded, _
            tlVHFACCaptureTerminationTerm50)
    .VoltageRange = 1.024
    .SampleRate.Enable = True
    .SampleRate.Value = 80000000#
```

```
        .LowPass.Bypass = True
    End With
' Reset the Peak Detector Circuit
    TheHdw.VHFACCapture.Pins(CapPin).PeakDetect.Reset
    ' Wait 1 us per the hardware spec for reset before measurement
    TheHdw.Wait (0.000001)
' SiteDouble variables for measurements (assuming single-pin capture):
'   Meas is SiteDouble to capture data
'   SumMeas is SiteDouble to accumulate sum of captured data per site
'   AvgMeas is SiteDouble for average at each site
Dim Meas As New SiteDouble
Dim SumMeas As New SiteDouble
Dim AvgMeas As New SiteDouble
' Perform "NumSamples" number of Peak Measurements
    For i = 0 To NumSamples - 1
        Meas = TheHdw.VHFACCapture.Pins(CapPin).PeakDetect.Measure
        SumMeas = SumMeas.Add(Meas)
' Average measurement at each site
    AvgMeas = SumMeas.Divide(NumSamples)
```

In this example, the syntax that begins with the Call keyword is used to program the VHFAC Capture to be in “Peak Detect” mode:

```
Call .SetMode(tlVHFACCaptureCaptureTypePeakDetect, _
```

The remaining capture parameters are programmed in the rest of the call statement:

```
                tlVHFACCaptureInputModeSingleEnded, _
                tlVHFACCaptureTerminationTerm50)
        .VoltageRange = 1.024
        .SampleRate.Enable = True
        .SampleRate.Value = 80000000#
        .LowPass.Bypass = True
```

Before the measurement, you must reset the peak detect circuit:

```
TheHdw.VHFACCapture.Pins(CapPin).PeakDetect.Reset
```

One microsecond must elapse for the reset to complete:

```
TheHdw.Wait (0.000001)
```

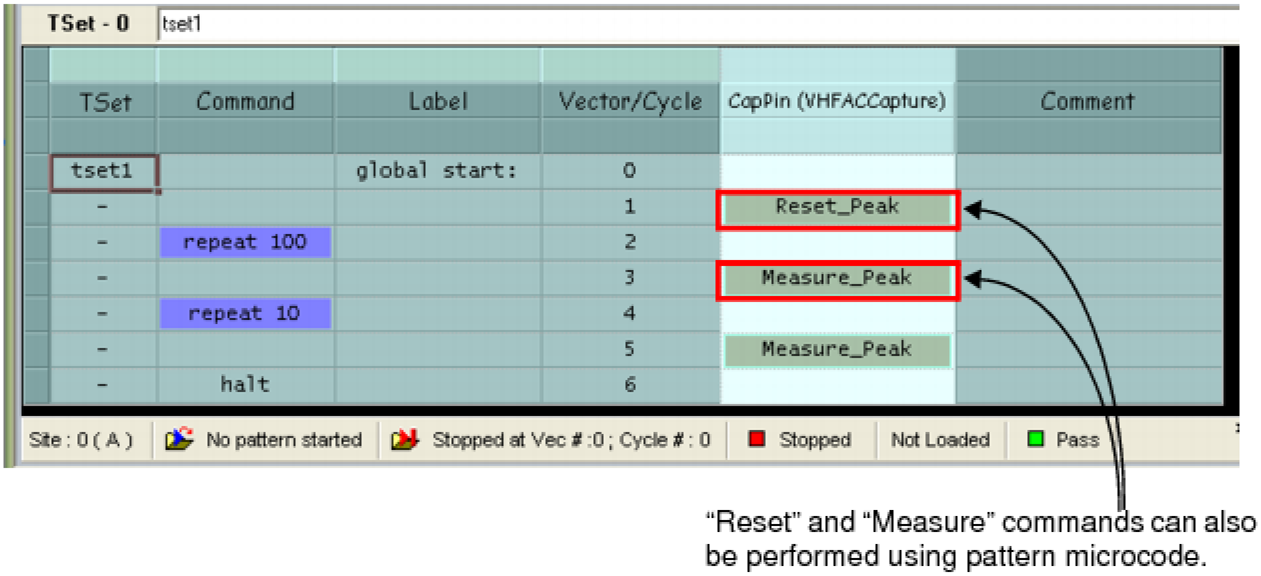
Now you can perform the measurement, capturing the results in the PinListData variable “Meas” and accumulating results in SumMeas:

```
Meas = TheHdw.VHFACCapture.Pins(CapPin).PeakDetect.Measure
SumMeas = SumMeas.Add(Meas
```

Finally, once all measurements ("NumSamples" of them) are complete, you calculate the average of the measurements at each site:

AvgMeas = SumMeas.Divide(NumSamples)

Using Pattern Microcode to Program Peak Detect




Using the Pattern Editor to Set Microcodes for Peak Detect

Here is an example:

TheHdw.VHFACCapture.Pins(CapThe fPin).PeakDetect.StartAddress = 50

TheHdw.Digital.Patterns.Pat("pattern.pat").Load

TheHdw.Digital.Patterns.Pat("pattern.pat").Start ("start")

Set Meas = TheHdw.VHFACCapture.Pins(CapPin).PeakDetect.ReadMem(50, 0)

In this example, you start by explicitly setting the StartAddress for measurements to be made by the pattern:

TheHdw.VHFACCapture.Pins(CapThe fPin).PeakDetect.StartAddress = 50

Note that StartAddress can be programmed using a PSet.

You then load and start the pattern:

TheHdw.Digital.Patterns.Pat("pattern.pat").Load

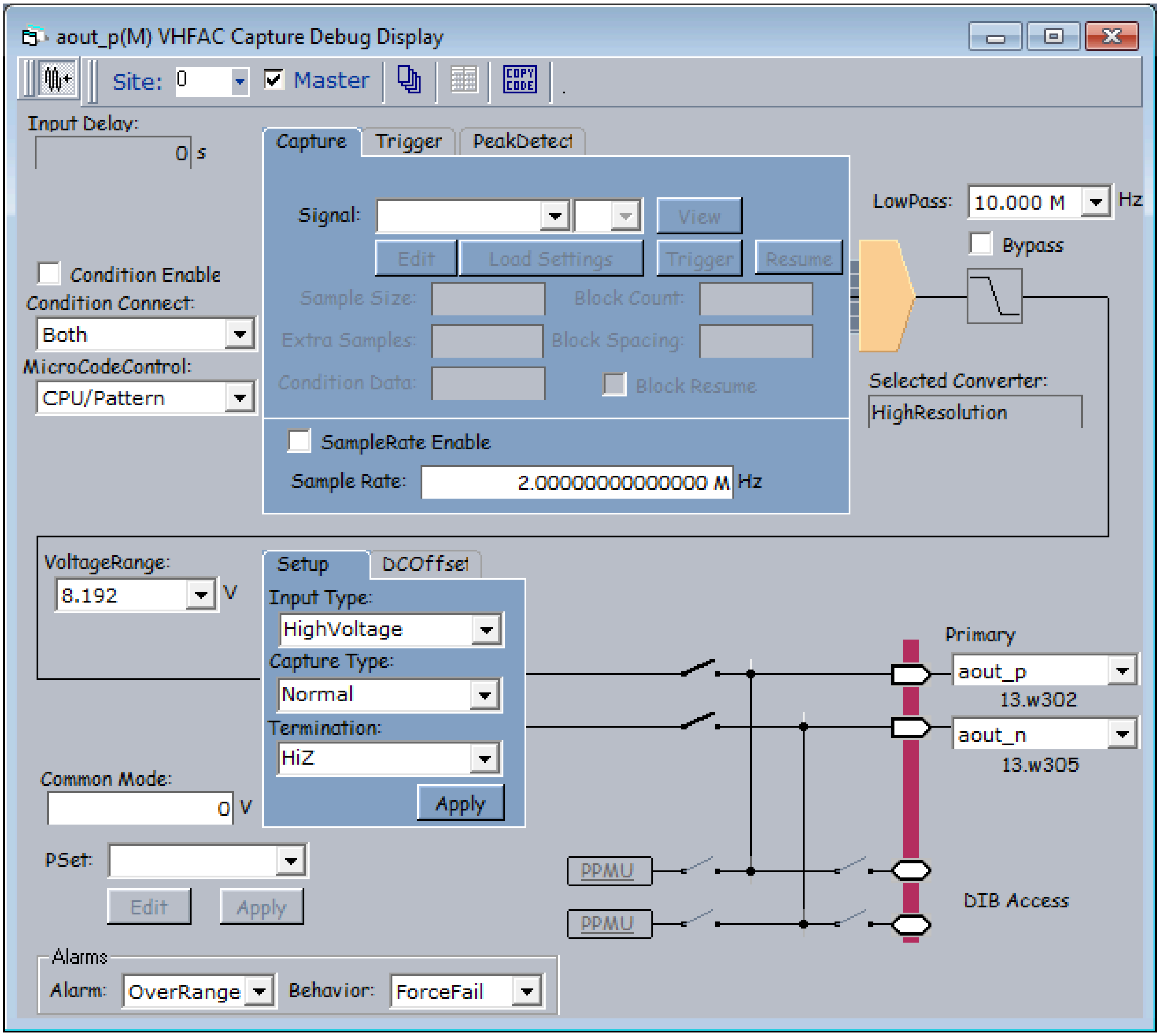
TheHdw.Digital.Patterns.Pat("pattern.pat").Start ("start")

Finally, you read the measured results and store them into the variable "Meas":

Meas = TheHdw.VHFACCapture.Pins(CapPin).PeakDetect.ReadMem(50, 0)

Performing Real Time Measurements in the Debug Display

Once the Peak Detect circuit is programmed, you can see the real time results in the VHFAC Capture Debug Display as shown in the figure. The Reset and Measure buttons are used for control, and real time results are shown in the Peak Detect Value field.




Viewing Results in the VHFAC Capture Debug Display

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VHFAC Capture Theory of Operation

VHFAC Capture Theory of Operation

This section provides conceptual and reference information about the VHFAC Capture hardware, features, and settings. The following topics are available:

- VHFAC Capture Theory of Operation
- VHFAC Capture Reset Values

For information on VHFAC Capture instrument performance specifications and standards, see VHFAC Specifications.

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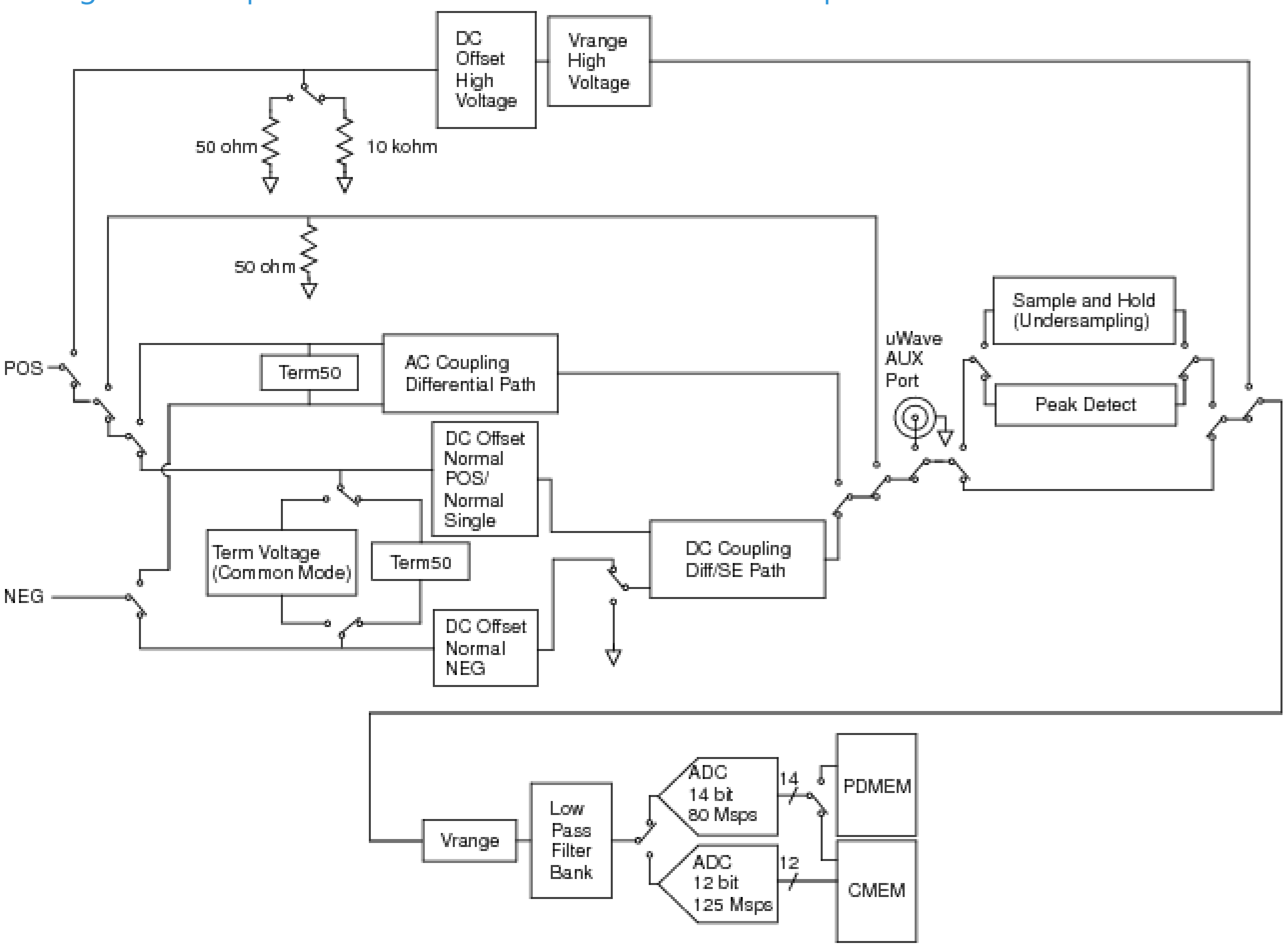
VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation

## VHFAC Capture Theory of Operation

The figure below provides a detailed view of the VHFAC capture instrument.




## VHFAC Capture Instrument Components

VHFAC Capture operation is described in the following topics:

- Input Capabilities

- Coupling and Offset
- Capture Type Stages
- Voltage Ranging
- Programmable Low-Pass Filter
- Capture Memory and Peak Detect Memory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VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Input Capabilities

Input Capabilities

An input signal enters the VHFAC Capture instrument as an analog signal from the DUT through the standard Pogo input path. The signal is an arbitrary waveform that must have an amplitude less than the programmed voltage range and frequency components less than the instrument's bandwidth to be accurately captured. The input signal can alternatively be captured from the AUX port (auxiliary output path) rather than from the DUT.

The VHFAC Capture can select single-ended or differential input. Differential input can be AC or DC-coupled. If single-ended input is selected, it can be:

| | |
|---|---|
| Single-Ended - Normal | Uses the normal single-ended path in hardware. Offset can be removed from the input signal. Termination should be 50 ohms in this mode. |
| Single-Ended - Direct | Uses the single-ended path in hardware, but bypasses all offset circuitry. Termination should be 50 ohms in this mode |
| Single-Ended - High V | Uses the high voltage single-ended path in hardware. Offset can be removed from the input signal. Termination should be high impedance or 50 ohms in this mode. |

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VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Coupling and Offset

Coupling and Offset

The coupling and offset circuitry of the VHFAC Capture allows you to achieve the smallest AC differential signal by removing differential offset from the capture signal. This is advantageous because by removing the differential offset of the signal you can get the optimal dynamic range of the capture range selected. Additionally, common mode offset can be removed from the signal (if necessary) to ensure that the signal on each pin with respect to VHFAC ground does not exceed the instrument's compliance.

The captured signal can be DC or AC coupled. When DC coupled, the signal's AC and DC components are passed. The advantage of being DC coupled is that the instrument settling time is less because the high-pass filter used for AC coupling is bypassed. When AC coupling is applied, the DC component of the signal is blocked, and only the AC components are passed.

The DC offset removal circuitry can be applied to the positive pin, the negative pin, or both. When the DC offset removal is applied to both pins, since there is only one offset register, the same voltage offset is applied to both.

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VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Capture Type Stages

Capture Type Stages

After passing through the coupling and offset circuitry, the signal enters one of three user selectable capture type stages: normal, undersampling, or peak detect.

Valid VHFAC Capture ranges are listed in the table below. The combination of Capture Type, Input Type, and Termination can be configured as Input Path.

VHFAC Capture Ranges

| Input Path | | | |
|---|---|---|---|
| Capture Type | Input Type | Termination | Voltage Range |
| Normal | DC Differential | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096 |
| | DC Differential | Voltage | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048 |
| | AC Differential | 50 ohms | 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096 |
| | Single-Ended Normal | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048 |
| | Single-Ended Direct | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024 |
| | Single-Ended HighVoltage | HiZ | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192 |

| | | | |
|---|---|---|---|
| | Single-Ended HighVoltage | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048, 4.096, 8.192 |
| Undersampling | DC Differential | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048 |
| | DC Differential | Voltage | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024 |
| | AC Differential | 50 ohms | 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024, 2.048 |
| | Single-Ended Normal | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512, 1.024 |
| | Single-Ended Direct | 50 ohms | 0.008, 0.016, 0.032, 0.064, 0.128, 0.256, 0.512 |
| Peak Detect | DC Differential | 50 ohms | 4.096 |
| | DC Differential | Voltage | 2.048 |
| | AC Differential | 50 ohms | 4.096 |
| | Single-Ended Normal | 50 ohms | 2.048 |
| | Single-Ended Direct | 50 ohms | 1.024 |

VHFAC Capture Supported Modes

| Capture Type | Input Mode | Termination | DC Baseline Removal Debug Display Field | Sample Rate | Voltage Range |
|---|---|---|---|---|---|
| Normal | AC Differential | Term50 | None | 3.434k - 125M | 0.016 - 4.096[1] |
| | DC Differential | Term50 | DC Offset Normal Pos DC Offset Normal Pos | 3.434k - 125M | 0.008 - 4.096[1] |
| | | Voltage | Common Mode | 3.434k - 125M | 0.008 - 2.048[1] |
| | SingleEnded | Term50 | DC Offset Normal Single | 3.434k - 125M | 0.008 - 2.048 |
| | HighVoltage | Term50 | DC Offset Normal High Voltage | 3.434k - 80M | 0.008 - 8.192 |

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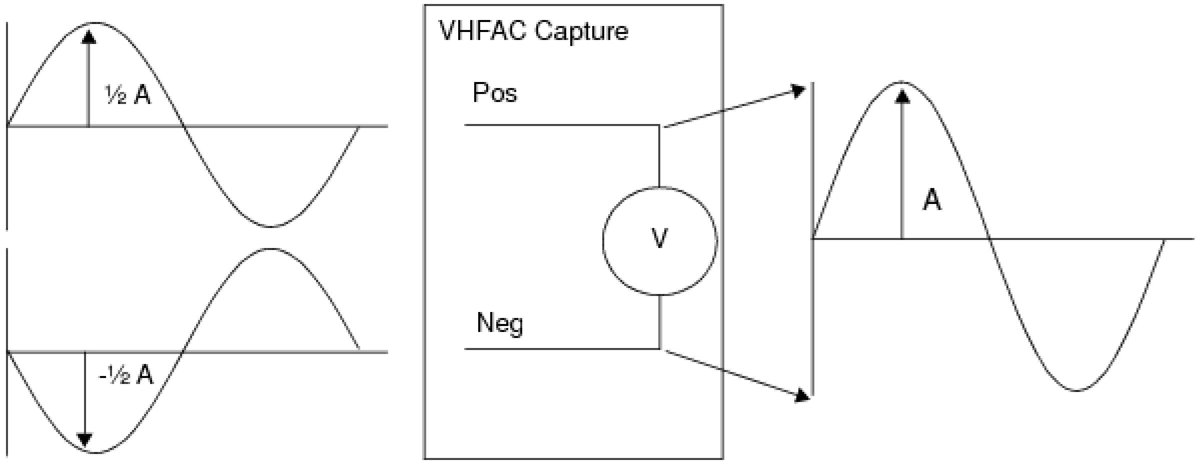
|---|---|---|---|---|---|
| | | HiZ | DC Offset Normal High Voltage | 3.434k - 80M | 0.008 - 8.192 |
| | Direct | Term50 | None | 3.434k - 125M | 0.008 - 1.024 |
| Under-sampling | AC Differential | Term50 | None | 3.434k - 20M | 0.016 - 2.048[1] |
| | DC Differential | Term50 | DC Offset Normal Pos DC Offset Normal Pos | 3.434k - 20M | 0.008 - 2.048[1] |
| | | Voltage | Common Mode | 3.434k - 20M | 0.008 - 1.024[1] |
| | SingleEnded | Term50 | DC Offset Normal Single | 3.434k - 20M | 0.008 - 1.024 |
| | Direct | Term50 | None | 3.434k - 20M | 0.008 - 0.512 |
| PeakDetect | AC Differential | Term50 | None | 3.434k - 80M | 4.096[1] |
| | DC Differential | Term50 | None | 3.434k - 80M | 4.096[1] |
| | | Voltage | None | 3.434k - 80M | 2.048[1] |
| | SingleEnded | Term50 | None | 3.434k - 80M | 2.048 |
| | Direct | Term50 | None | 3.434k - 80M | 1.024 |

1 Differential voltage specifications assume a differential voltage across the two inputs. For example, in the 2.048 V range, the voltage at the high input can be 1.024 V peak and the voltage at the low input can be -1.024 V peak. This yields a differential voltage of 2.048 V across the high and low inputs.

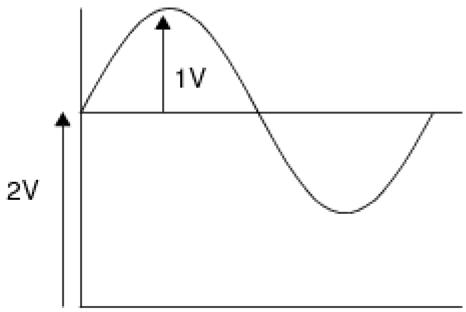
For example, in the figure, the input signal is a balanced differential sine wave with amplitude A. The maximum and minimum voltage ever presented between the positive and negative is at the peaks, A volts. The capture range must, therefore, be set to a value greater than or equal to A. This is not the same as the peak-to-peak voltage of the sine wave, which is 2*A. Peak-to-peak refers to the maximum voltage difference encountered over time.




Differential Signal Capture

When dealing with ranging issues, the instrument is only concerned with the instantaneous voltage. The maximum Vpp of a captured sine wave is only twice the range voltage when the signal has no differential offset.

For example, the following figure shows that the differential signal needs a range of at least 4.096V to avoid saturating the capture (assuming DC coupling and the baseline circuit is not used). The 1.024V or even 2.048V ranges are too small for this signal because the combination of the AC and DC components creates a signal that has a maximum instantaneous voltage differential of 3V.




Signal with 2V DC Offset

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# VhfacCapture_theory.2.06

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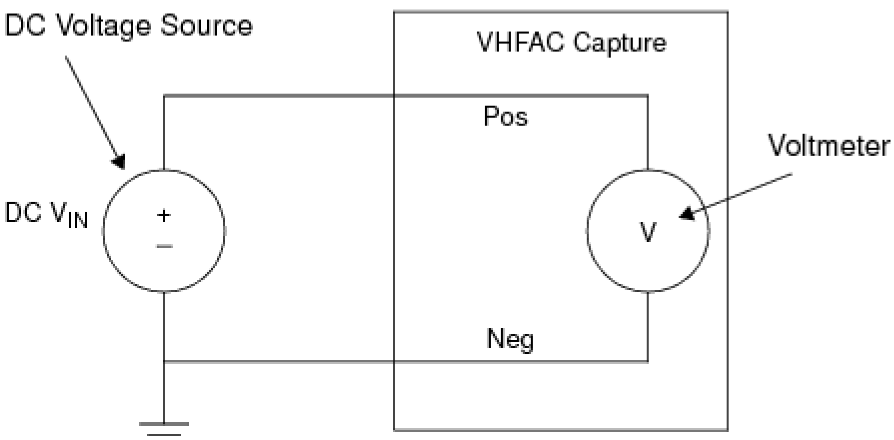
VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Voltage Ranging

Voltage Ranging

After the undersampling and peak detect stage, the signal enters the voltage ranging circuitry. This circuitry is used to scale the signal to ensure that it fits within the dynamic range of the analog-to-digital converters. The range specifies the maximum instantaneous voltage difference that can be applied between the positive and negative input pins.

This figure shows that voltage ranging for the VHFAC Capture can be schematically represented as a voltmeter, where $V_{in}$ is the voltage applied to the voltmeter.




VHFAC Capture Comparison to a Voltmeter

The measurement in this example does not qualify as a single-ended measurement because every measurement must be made between Positive and Negative. Therefore, all measurements are defined as a differential voltage. Furthermore, peak-to-peak voltage (Vpp), and to a lesser extent peak voltage (Vpk), have no meaning here because they refer to the properties of a signal, not the measuring instrument.

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# VhfacCapture_theory.2.07

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VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Programmable Low-Pass Filter

Programmable Low-Pass Filter

The scaled input signal passes through a bank of user selectable low-pass filters to attenuate band frequency components that would result in aliased signal components. Valid filter settings are: bypass, 500 kHz, 2 MHz, 6.1 MHz, 10 MHz, 20MHz, 40MHz, and 62.5MHz.

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# VhfacCapture_theory.2.08

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VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Analog-to-Digital Converter

Analog-to-Digital Converter

The filtered analog signal is then converted to a digital signal using the “high resolution” (14bit, 80 Msps) or the “high speed” (12 bit,125 Msps) A/D converter. The choice of converter is determined by IG-XL based on the user programmed sample rate.

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# VhfacCapture_theory.2.09

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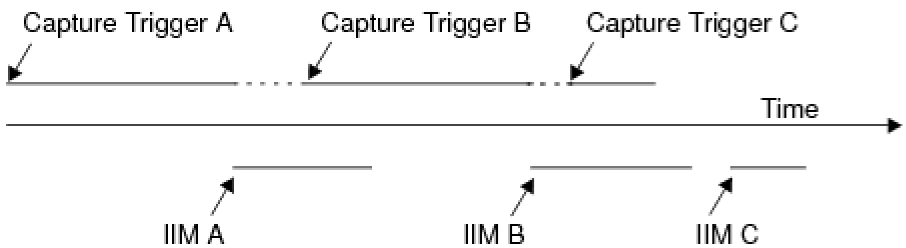
VHFAC Capture Theory of Operation : VHFAC Capture Theory of Operation : Capture Memory and Peak Detect Memory

Capture Memory and Peak Detect Memory

Digital values from the ADC are routed to a 1M capture memory or a 256 sample peak detect memory. The VHFAC Capture memory (CMEM) uses an instrument initiated move (IIM) process to automatically move the captured data from the instrument's capture memory to the support board's DSP modules for signal analysis. Each capture begins where the previous capture ended. As soon as a capture completes, the next capture can begin.

The figure shows three consecutive captures. As soon as the first capture completes, it is automatically moved through IIM for processing, and the second capture begins while the first is moving. The third capture then begins and completes while the second is still moving. As soon as a capture is done moving, its space in CMEM is freed.




Capture Sequence and Instrument Initiated Move (IIM)

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# VhfacCapture_theory.2.10

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VHFAC Capture Theory of Operation : VHFAC Capture Reset Values

VHFAC Capture Reset Values

This table lists the local reset values for the VHFAC Capture. Pre-job reset, post-job reset, power-up reset, and user reset all result in the same state, unless otherwise noted.

VHFAC Capture Instrument Reset Values

| Parameter | Value |
|---|---|
| Capture Type | Normal |
| Input Type | Single-Ended HighVoltage |
| Termination | HiZ |
| Voltage Range | 8.192 |
| LowPass | 2 MHz |
| LowPass Bypass | False |
| Sample Rate | 2 MHz |
| Sample Rate Enable | False |
| InputDelay | 0 |
| Offset Single-Ended Normal | 0 |
| Offset Single-Ended HighVoltage | 0 |
| Offset DCDifferential Positive | 0 |
| Offset DCDifferential Negative | 0 |
| CommonModeDC | 0 |

| Connections | Disconnected |
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